

Chapter 1

Introduction

The performance and energy of a modern computer are increasingly limited not by how fast it can compute, but by how much data it must move between where information is stored and where it is processed [1, 2]. Two trends have made this limit acute. First, the dominant workloads have shifted toward large machine-learning models and continuous streams of sensor data, both of which are defined by the sheer number of memory accesses they require [3]. Second, the transistor scaling that once absorbed these costs has slowed, so the energy of data movement is no longer hidden by each new process generation. The response, across industry and research alike, has been to bring memory and computation physically closer together, and—more radically—to let the dynamics of the physical device itself perform part of the computation. The biological synapse is the canonical example of the latter idea: a single element that transmits a signal, modifies its strength according to recent history, and stores that history across multiple timescales [4].

This thesis examines organic memristive devices built from ion-containing polymer composites and explores how their internal dynamics can serve as exactly this kind of computational resource. The work lies at the

intersection of solid-state electronics, polymer physical chemistry, and computational neuroscience: a two-terminal device must behave like an electrical circuit element, its state must be set by ion motion in a soft host, and its usefulness is judged by whether that state provides the fading memory needed for temporal computation.

1.1 The Crisis in Modern Computing

Modern digital computers rest on an architecture that has been extraordinarily successful at executing arbitrary programs on general-purpose hardware. That success, however, was built on an assumption that no longer holds: that the cost of a computation is essentially the cost of its arithmetic. On contemporary hardware the dominant cost is instead the transport of operands to and from memory, and simply making transistors smaller and clocks faster no longer addresses it [1, 5]. The practical limits are therefore architectural as well as technological: the *memory wall*, the *energy wall*, and the slowing of classical transistor scaling all originate in the separation between stored data and the circuitry that acts on it [2, 6].

1.1.1 Von Neumann Architecture and Its Limitations

The architectural model of modern digital computers is the *stored-program* concept set out by von Neumann in 1945 [7]: instructions and data share one memory and are fetched word by word by a central processing unit across a connecting bus. This is what makes a single machine general-purpose, and engineering refinements (registers, caches, branch prediction, memory controllers) have left the topology essentially unchanged. The same topology is the source of the limitation. Because the processor and the memory are *physically distinct* subsystems, every instruction

fetch, operand load, and result write-back must cross the channel between them; no arithmetic can begin before its operands arrive, and no result is preserved until it is moved back. This is the *processor-memory dichotomy*.

Because only a bounded amount of data can cross that channel per cycle—the *von Neumann bottleneck* [8]—performance is set by the slowest interface rather than by the arithmetic. Pipelining, multi-level caches, speculation, prefetching, and out-of-order issue all mitigate this, but they preserve the separation between computation and storage, so the bottleneck is shifted rather than removed. Wulf and McKee [6] named the resulting imbalance the *memory wall*: with processor speed long outpacing DRAM latency and bandwidth, an ever-larger share of each cycle is spent waiting on memory. Modern microprocessors accordingly devote much of their silicon area, and an even larger fraction of their power, to caches, prefetch logic, memory controllers, and interconnect—hardware whose sole purpose is to keep the arithmetic units supplied with data [5]. The bandwidth and energy aspects of this imbalance are summarised in fig. 1.1.

In 2014, Horowitz [1] provided a comparison of costs for elementary operations in a representative CMOS process where an addition of 32-bit integers costs on the order of 0.1 pJ, but an SRAM access can cost tens of picojoules, while a DRAM access could cost several hundred picojoules to nanojoules (table 1.1). These per-operation figures matter only in proportion to how often each operation is performed, and this is exactly where modern workloads make the asymmetry decisive. A single inference pass through a large neural network performs billions of multiply–accumulate operations, each of which must fetch a stored weight before it can do the comparatively cheap arithmetic; training repeats this over enormous datasets. The energy budget is therefore spent overwhelmingly on operand transport rather than on the arithmetic it enables. This is the *energy wall*, which comes from the parasitic capacitance of interconnects

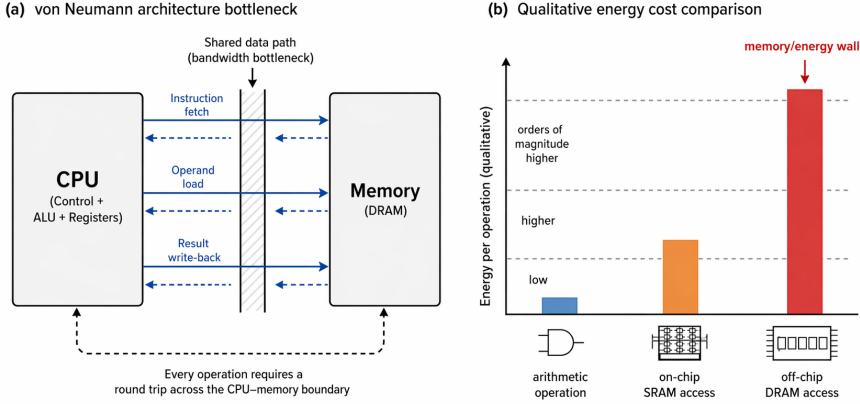


Figure 1.1: Schematic of the von Neumann architecture and the associated bandwidth and energy costs of data movement. (a) In the stored-program paradigm, the central processing unit and the memory unit are physically separated and communicate through a shared bus, so that every instruction fetch and every operand access must traverse that bus. (b) The energy cost of moving a word across this boundary exceeds, by orders of magnitude, the energy cost of the arithmetic operation performed on it, and this asymmetry has increased as technology has scaled [1].

and from the much greater total wire length of a memory subsystem than of an arithmetic unit. While computation and storage remain spatially separated, the architecture is limited by the physics of moving charge through wires.

For decades these inefficiencies were hidden by silicon scaling: Moore’s doubling of transistor counts [9] came paired with Dennard scaling [10], so each generation delivered more transistors that were also faster and cooler, and the memory and energy walls did not yet bind. Dennard scaling broke down around 2005, when leakage currents no longer allowed proportional voltage scaling [2]; transistor counts kept rising, but performance and energy gains slowed and power density became the limiting constraint—the *post-Dennard* or *dark-silicon* regime [11]. A new process node can therefore no longer hide the cost of the von Neumann model, and the problem must be confronted at the architectural and materials level. The pressure has

Table 1.1: Approximate energy cost per operation in a representative 45 nm CMOS process, after Horowitz [1]. Memory accesses cost between one and four orders of magnitude more energy than the arithmetic they enable, and the asymmetry widens with the size of the memory addressed and with the distance the data has to travel.

Operation	Energy	Class
8-bit integer add	0.03 pJ	Arithmetic
32-bit integer add	0.1 pJ	Arithmetic
32-bit floating-point add	0.9 pJ	Arithmetic
32-bit floating-point multiply	3.7 pJ	Arithmetic
Register-file read	0.1 pJ	On-chip storage
SRAM read (8 kB)	5 pJ	On-chip storage
SRAM read (1 MB)	100 pJ	On-chip storage
Cross-chip wire (~ 10 mm)	100 pJ	Interconnect
DRAM read	1.3 nJ	Off-chip storage

only grown as workloads shift toward machine-learning kernels dominated by repeated matrix operations on memory-resident parameters [3]: data movement worsens exactly as scaling stops compensating for it. Hennessy and Patterson [2] read this as an inflection point, arguing that the next stage of computer architecture will lean on domain-specific hardware in which memory and computation are brought much closer together.

1.1.2 In-Memory and Event-Driven Computing Paradigms

Two responses to this separation have emerged. The first keeps the digital workload largely intact but performs operations directly within, or very close to, the memory array; this route is described as *in-memory*, *near-memory*, or *processing-in-memory* computing. The second uses temporally structured event streams and treats the intrinsic dynamics of the physical device as part of the computation itself; this route is known as *event-driven* or *temporal* computing. The two responses, set against the von Neumann baseline they address, are summarised in fig. 1.2.

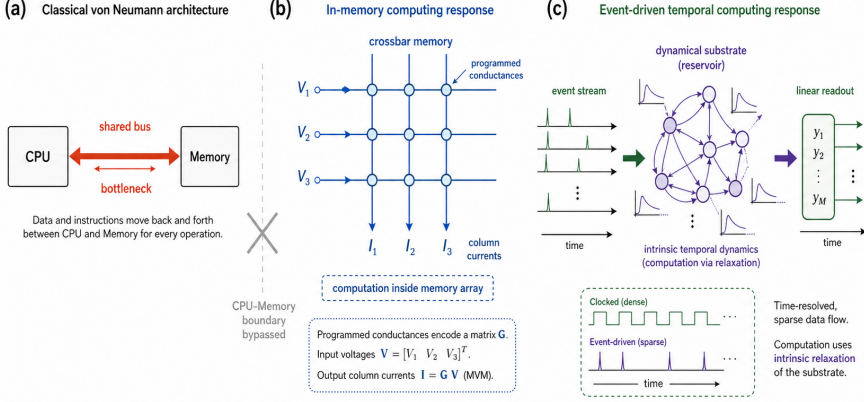


Figure 1.2: Two architectural responses to the von Neumann bottleneck. (a) In the classical stored-program architecture, every operation traverses a CPU–memory bus that fixes the bandwidth and energy ceilings of the system. (b) In the in-memory paradigm, computation is performed inside the memory array: a resistive crossbar with programmed conductances executes an analogue matrix–vector multiplication by Ohm’s and Kirchhoff’s laws, so the dot product no longer crosses the CPU–memory boundary [12, 13]. (c) In the event-driven paradigm, dense clocked data are replaced by a sparse stream of asynchronous events that drive a dynamical substrate whose intrinsic relaxation is itself the computation; only a linear readout layer downstream is trained [14–16].

In-memory computing bypasses the classical *fetch–compute–write-back* cycle by executing operations directly inside the memory array. This is demonstrated by implementing analogue matrix–vector multiplication using a resistive crossbar architecture, in three steps:

1. Each memory cell contains a conductance value that reflects a matrix entry.
2. Voltages are applied across the rows, representing the input vector.
3. The resulting column currents represent the sum of products, in accordance with Kirchhoff’s laws.

This changes the array from a passive storage device into an active computing tool. Demonstrations using metal-oxide resistive memories [12], phase-change materials [17, 18], and organic ionic devices [19–21] have achieved matrix–vector operations with energy efficiencies that surpass those of traditional digital accelerators, and similar principles can be extended to convolutional [22] and attention-like processes (the matrix-multiplication kernels at the core of transformer architectures [23]).

At the device level, however, this version of in-memory computing still requires memory properties. Each cell should hold a quasi-static conductance that encodes a trained weight, and which should be set once, read many times, and remain stable enough to meet the application’s numerical precision requirements. Non-volatility over seconds to years becomes a design goal.

Event-driven computing is motivated by the fact that many signals of interest carry information in time. Natural sounds, visual transients, motor commands, biomedical waveforms, and streaming sensor outputs are all non-static examples. For such signals, the computational model should be a system that receives events in continuous time and evolves in response, because information is not only present in the amplitude values but also in the waveform’s evolution and timing. At the algorithmic level, this is the domain of spiking neural networks [24]. At the circuit level, it corresponds to event-driven asynchronous logic. At the device level, which is the level of this thesis, it requires materials whose intrinsic relaxation dynamics are matched to the signals they process.

This framework mirrors biological nervous systems, which efficiently handle complex sensorimotor and cognitive tasks while consuming much less power than artificial systems. This efficiency partly results from architectural choices that diverge significantly from the von Neumann model. Neurons share information through sparse events, and synapses pass signals that change depending on recent activity. Information is

stored and altered through slow, gradual electrochemical processes, in which the ionic medium actively participates in computation. Although directly translating the brain into silicon isn't practical or useful, several principles do carry over: sparse-event coding [25], the use of slow device physics as a fundamental computing element, and viewing device heterogeneity as a feature for functional diversity rather than a flaw. These principles form the basis of neuromorphic engineering [26].

Boyd and Chua [27] showed that any time-invariant nonlinear operator with *fading memory*, in an appropriate function space, can be approximated by a finite Volterra series (a power-series expansion in time-shifted inputs). This means that a system whose current state retains a decaying record of past inputs can perform various temporal computations when paired with a trainable linear readout. This concept was later formalised through algorithms such as Jaeger's *echo state network* [14, 28] and Maass, Natschläger, and Markram's *liquid state machine* [15], which were combined under the umbrella of *reservoir computing* [29, 30]. In this framework, an untrained dynamical system—the reservoir—is driven by the input stream, with only the readout being trained. When the reservoir is realised in a physical substrate, the approach is known as *physical reservoir computing* [16], with implementations in optical, mechanical, spintronic, volatile-memristive, and organic electrochemical systems [31–34].

This redefines how a working device should be evaluated. If the goal is to maintain a trained weight permanently, a device that can't retain memory for long periods will be considered unsuitable. However, in a reservoir architecture, this finite memory retention *is* what defines the period during which previous inputs influence the current output, a concept referred to as the fading-memory timescale. Variability between devices can also play a different role: while it's a disadvantage for weight storage, a controlled variation in intrinsic timescales can enable a common readout

to interpret temporal structures over a wider range than any individual device could achieve [16].

1.2 The Biological Synapse as the Engineering Template

In neural circuits, a single physical element both transmits a signal and stores part of the history that shapes future transmissions: the synapse. The same molecular event that relays information from one neuron to the next also alters its future strength, so signalling and memory share a substrate. The device target is therefore synaptic signalling and plasticity rather than full neuron emulation.

1.2.1 Neural Signal Transmission: A Brief Overview

A neuron is a polarised cell specialised to receive, integrate, and transmit electrical signals [35]. Dendrites collect inputs from other cells, the soma integrates them, and when the membrane potential at the axon hillock crosses a threshold the axon propagates an outgoing signal whose terminals release a chemical messenger onto downstream cells. A mature animal neuron typically receives 10^3 – 10^4 such contacts [36], and each is a *synapse*. How a neuron computes is therefore governed far more by the number and strength of its synapses, and by how they change over time, than by the soma itself.

The axonal signal is the *action potential*: a stereotyped, millisecond-scale depolarisation that is all-or-nothing, so that information is carried not in its amplitude but in the timing, rate, and coincidence of spike trains [37, 38]. Neural computation is in this sense event-driven (section 1.1.2) and arises from the pattern of signals arriving at many synapses. The neuron's

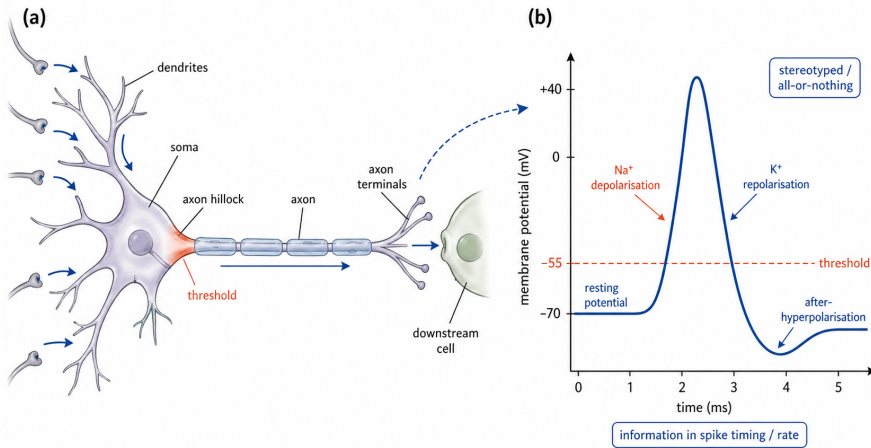


Figure 1.3: Anatomy of a neuron and the action-potential waveform. (a) A neuron collects synaptic inputs through its dendrites, integrates them at the soma, and fires an outgoing electrical event when the membrane potential at the axon hillock crosses a threshold; the resulting waveform propagates unidirectionally along the axon to the synaptic terminals, where it triggers chemical transmission onto downstream cells (the zoom shown in fig. 1.4). (b) The action potential is a stereotyped millisecond-scale depolarisation produced by the opening of voltage-gated Na^+ channels, followed by a K^+ -mediated repolarisation and a brief after-hyperpolarisation before the return to the resting potential; its all-or-nothing character means that information about the input is carried by the timing and rate of these events rather than by their amplitude, which is the cellular basis of the event-driven view of neural computation introduced in section 1.1.2.

anatomy and the action potential waveform are summarised in fig. 1.3.

Synaptic transmission is largely chemical. An action potential arriving at the terminal opens voltage-gated Ca^{2+} channels, and the Ca^{2+} influx triggers neurotransmitter-filled vesicles (glutamate at excitatory cortical synapses) to fuse with the presynaptic membrane [39]. The transmitter diffuses across the ~ 20 nm cleft and binds postsynaptic receptors [36], producing an excitatory or inhibitory postsynaptic current (EPSC or IPSC) whose amplitude and kinetics define the *strength* of the synapse at that moment. Crucially, the same molecular machinery that carries the

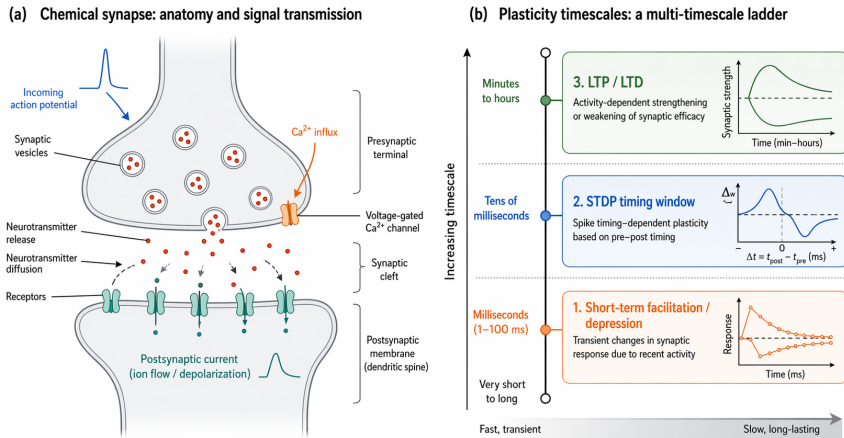


Figure 1.4: Anatomy and plasticity regimes of the chemical synapse. (a) A presynaptic terminal releases neurotransmitter vesicles into the synaptic cleft in response to a calcium-triggered action potential, and postsynaptic receptors convert the released signal into an excitatory postsynaptic current. (b) The same synapse supports a graded hierarchy of plasticity regimes — short-term facilitation and depression on millisecond timescales, spike-timing-dependent plasticity on tens of milliseconds, and long-term potentiation and depression on timescales from minutes to hours — each associated with distinct microscopic mechanisms.

signal also sets the strength of future transmissions.

The synapse is a transmission element, a modulator, and a memory element at the same time: it transmits a signal, adjusts the amplitude and time course of that signal, and stores lasting changes produced by previous activity. This co-localisation of transmission, modulation, and memory makes the synapse, more than the neuron as a whole, the most useful biological template for post-von-Neumann electronic devices. The relevant anatomy and plasticity regimes are summarised in fig. 1.4.

1.2.2 Synaptic Plasticity: The Physical Basis of Learning

Experience leaves a physical trace in the nervous system, stored mainly at synapses. Hebb’s 1949 postulate—that a presynaptic cell which repeatedly helps fire a postsynaptic cell strengthens its connection to it [40] (“cells that fire together, wire together”)—captures two features that matter for hardware: the rule is *local* to the two cells a synapse joins, and *sensitive to correlation*. Synaptic plasticity implements Hebbian or Hebbian-like behaviour across several timescales.

On the shortest timescales, *short-term plasticity* [41, 42] makes the immediate strength of a synapse depend on its recent activity. *Facilitation* is a transient increase in EPSC amplitude when a second spike follows the first after a short delay, usually attributed to residual presynaptic Ca^{2+} ; *depression* is a transient decrease caused by vesicle depletion or receptor desensitisation. With time constants from tens of milliseconds to several seconds, short-term plasticity makes the synapse a nonlinear, history-dependent temporal filter, so that identical spike sequences produce different responses depending on their timing.

On longer timescales, *long-term potentiation* (LTP) provides persistent, activity-dependent strengthening. First demonstrated by Bliss and Lømo in the rabbit hippocampus [43] and later shown to last days to weeks [44, 45], hippocampal LTP relies on NMDA receptors that open only when the postsynaptic membrane is already depolarised (a voltage-dependent Mg^{2+} block [46, 47]), making them Ca^{2+} -permeable coincidence detectors; the resulting Ca^{2+} influx drives additional AMPA receptors into the postsynaptic membrane [48, 49]. The complementary *long-term depression* (LTD) [50, 51] weakens synapses under low-frequency or anti-correlated activity, principally through reverse AMPA-receptor trafficking [52]. Together, LTP and LTD give the bidirectional, long-lasting amplitude

1.2. THE BIOLOGICAL SYNAPSE AS THE ENGINEERING TEMPLATE

changes considered a primary physiological basis of learning.

Spanning these regimes, *spike-timing-dependent plasticity* (STDP), described in cortical pyramidal cells [53] and quantified in hippocampal cultures by Bi and Poo [54], makes the sign and magnitude of the change depend on the delay between pre- and postsynaptic spikes within a window of a few tens of milliseconds: a presynaptic spike before the postsynaptic one potentiates, the reverse depresses, and large delays in either direction produce no lasting change. The synapse thus holds a short memory of recent presynaptic activity, compares it with postsynaptic firing, and converts that timing relationship into a longer-lasting amplitude change.

Across these mechanisms the synapse is not a passive store of a single “weight” but operates over many timescales at once: milliseconds for individual spikes, tens to thousands of milliseconds for short-term facilitation and depression, and minutes to hours and beyond for LTP and LTD. Driven by partially overlapping but distinct molecular mechanisms, these layers led Abbott and Regehr [4] to argue that a synapse is better understood as a computational unit than as a memory store. For hardware, both the long-lasting and the decaying state changes matter.

1.2.3 Why Hardware Emulation of the Synapse?

The motivation for hardware synapses is directly derived from section 1.1. In traditional computing, a software neural network performs calculations on data retrieved from a memory hierarchy physically separate from the arithmetic units. Each forward pass incurs data movement and energy costs as described earlier, and these costs increase with the number of simulated synapses. A typical human brain contains approximately 10^{14} – 10^{15} synapses [55] and consumes about 20 W [56]. A standard von Neumann architecture that consumes orders of magnitude more energy cannot easily bridge this gap through improved design alone, since

the cost of fetching a weight before multiplying it by an activation is limited by interconnect physics. Thus, in nature, the synapse exemplifies the most compact, low-power form of co-localised analogue memory and computation, making it an optimal physical template. The order-of-magnitude envelopes of these implementation regimes are compared in fig. 1.5.

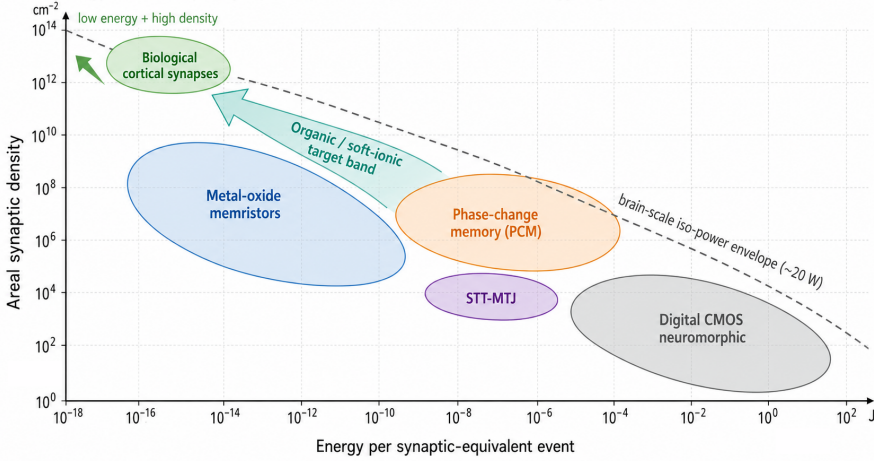


Figure 1.5: Approximate energy and density envelopes of synaptic-equivalent operations across implementation regimes. Each cluster represents a class of physical realisation: biological cortical synapses, digital CMOS neuromorphic implementations in which a synapse occupies an SRAM cell plus an arithmetic unit, the inorganic memristor families surveyed in section 1.4, and the soft-ionic composite target region forward-referenced to section 1.6. The diagonal reference line marks the brain-scale iso-power envelope (~ 20 W for $\sim 10^{14}$ – 10^{15} synapses) introduced at the start of section 1.2.3. Numerical bounds are conservative orders of magnitude drawn from the reviews cited in the surrounding text; individual devices may sit outside the envelopes shown.

The research programme built around this idea is *neuromorphic engineering*, whose programmatic origin is usually attributed to Mead [26]. Its goal is not merely to accelerate neural algorithms but to build circuits in which device physics, topology, and dynamics themselves embody computational characteristics like those of the nervous system, and within it

1.2. THE BIOLOGICAL SYNAPSE AS THE ENGINEERING TEMPLATE

the synapse has become the main device focus.

A hardware synapse should satisfy several requirements:

- It should provide an analogue, multi-state conductance that can be set reversibly by electrical stimuli and read non-destructively with a small probe current.
- It should also be compact enough for large arrays, so that many thousands of elements can be addressed with a manageable number of input and output lines [25, 57].
- Its response should be nonlinear in a meaningful way, for example by displaying a pulse-number or pulse-amplitude threshold below which little change is induced [13, 57].
- Finally, its dynamics should match the signals it is meant to process, whose relevant timescales span from milliseconds to seconds and beyond.

Two of these requirements deserve emphasis. The first is *analogue*, multi-state conductance: a binary element stores one bit and would need many devices per connection to represent a graded weight, whereas an analogue element encodes several distinguishable states in one device, and for ionic devices the write energy can be low because the state is set by moving ions over short distances rather than charging large digital circuits [20, 21]. The second is *reversibility*: a device that can be potentiated but not reliably depressed cannot follow the bidirectional rules of section 1.2.2 and so cannot fully replicate synaptic learning [13]. The resulting specification—analogue, reversible, multi-state, low-energy conductance changes set by electrical pulses—is the one that motivates the devices studied in the rest of the thesis.

The biology of section 1.2.2 reveals that biological synapses incorporate quasi-stable memory via LTP and LTD, while providing temporal filter-

ing through short-term plasticity and STDP, allowing both functions to coexist within the same physical structure. A hardware system can then be useful in two complementary ways: as a persistent or quasi-persistent programmable weight (as seen in crossbar arrays for in-memory computing [13]), and as a dynamic fading-memory element. The second role is the primary focus of this thesis, following the reservoir-computing approach to event-driven temporal computing introduced in section 1.1.2 [16, 31–33].

This leads to a materials question. In biology, the timescales of short-term plasticity are determined by different processes such as Ca^{2+} buffering, vesicle recycling, receptor desensitisation, and other local mechanisms that vary across synapses [42]. A hardware alternative for this role should offer an analogous way to adjust its electrical timescale. Classical transistor-based memories do not provide that kind of analogue, history-dependent element; memristive devices are the natural two-terminal candidate.

1.3 The Memristor: Theory and History

A memristor or memristive device shows a current–voltage (i–v) response that depends on an internal state, which itself is influenced by the device’s previous electrical activity. The ideal form of this is the *memristor*; the broader experimental category is the family of *memristive devices*. The point needed here is simple but important: a device can be memristive even when its internal state relaxes after the input is removed. Volatility does not disqualify the device; it defines the fading-memory behaviour exploited later in the thesis.

1.3.1 Memristive Systems as History-Dependent Devices

A memristor is, intuitively, a resistor with memory: a two-terminal element whose resistance depends on the charge that has previously passed through it. Chua [58] introduced it in 1971 on symmetry grounds, completing the set of fundamental passive elements (resistor, capacitor, inductor) with a fourth that relates charge $q = \int i \, dt$ to flux linkage $\varphi = \int v \, dt$. It is described by $v = M(q) i$, where the *memristance* $M(q)$ has the units of resistance but, unlike an ordinary resistor, carries a record of the device's past.

If M is constant the element is an ordinary resistor; if M varies with q , the present response depends on the history of the device. Under periodic excitation this produces a *pinched hysteresis loop* in the i - v plane [59]: the loop passes through the origin (zero voltage gives zero current instantaneously) but its shape reflects the internal state left by prior stimulation. Pinched hysteresis is the standard experimental signature of memristive behaviour [60], and it is collected together with Chua's constitutive argument in fig. 1.6.

Few real devices satisfy the ideal relation $\varphi = \hat{\varphi}(q)$. The broader and more useful notion, introduced by Chua and Kang [61] in 1976, is the *memristive system*: an input-driven internal state $\mathbf{x}$ evolves under generally nonlinear dynamics and gates the instantaneous electrical response. For this thesis, the crucial allowance is that $\mathbf{x}$ may include relaxation terms that drive the device back toward equilibrium once the input is removed. A volatile device whose state decays is therefore still a memristive system, even though it is not an ideal memristor. This is the framework adopted for the polymer-composite devices reported here.

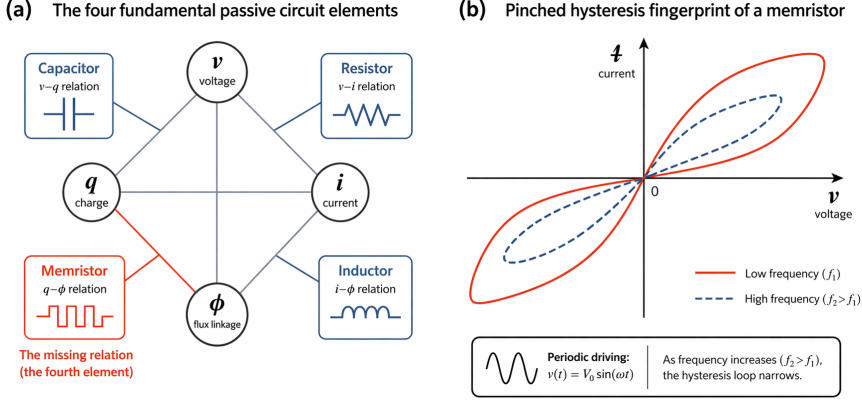


Figure 1.6: The memristor as a constitutive element and its experimental fingerprint. (a) The four fundamental two-terminal passive circuit elements, arranged by the pair of constitutive variables they relate: the resistor ($v-i$), the capacitor ($v-q$), the inductor ($i-\phi$), and the memristor ($q-\phi$), which completes the quadrant [58]. (b) A memristive device driven by a periodic voltage exhibits a pinched current–voltage hysteresis loop that narrows with increasing driving frequency and that passes through the origin at each zero-crossing; this loop is the canonical experimental fingerprint of memristive operation and is observed in all of the device families surveyed in the remainder of the chapter [59–61].

1.3.2 The First Physical Realisation: TiO_2 Memristor

For more than three decades after Chua’s proposal, the memristor remained mostly a theoretical construct. Thin-film metal-oxide resistive switches with hysteretic current–voltage characteristics had already been reported since the late 1960s [62–64], and many of them can now be recognised as memristive systems. At that time, however, they were not interpreted within Chua’s formalism. The field still lacked a standard physical device that connected the theoretical idea to a tangible thin-film structure.

In 2008, that connection was made possible by Strukov, Snider, Stewart

and Williams [65] at HP Laboratories. Their device consisted of a thin titanium dioxide (TiO_2) layer sandwiched between two platinum electrodes. Within the film, a reduced TiO_{2-x} region acted as an oxygen-deficient conductive layer, while the neighbouring TiO_2 region served as a resistive barrier. When a voltage was applied, oxygen vacancies moved, shifting the boundary between these two regions [66]. Given that the total resistance depends on the relative thickness of the conductive and resistive parts, the device resistance could then be modulated continuously by moving this internal boundary back and forth. The boundary position is the internal state, oxygen-vacancy drift updates it, and the resulting i - v response reproduces the pinched hysteresis predicted by the theory.

This result transformed the memristor from a formal circuit element into a concrete experimental object, thus giving life to a large body of work on resistive switching and memristive devices. A two-terminal resistance that can be programmed by electrical pulses, tuned over several conductance values, and read at low voltage is close to the analogue weight element required by neuromorphic hardware [67]. Crossbar demonstrations of in-memory matrix-vector multiplication followed soon after [12, 13]. The 2008 TiO_2 device is, however, only one mechanism—oxygen-vacancy drift in a transition-metal oxide—whereas the memristive formalism is far broader.

1.3.3 Classification of Memristive Mechanisms

In the field of device physics, the term *memristor* is usually considered a phenomenological concept, not a description of a particular microscopic mechanism. A device is considered "memristive" if its current-voltage response exhibits the pinched hysteresis that Chua described and if its internal state can be both set and read electrically. This can be achieved through many different physical processes, commonly grouped into broad

mechanistic families [13, 67].

These taxonomies combine two different classification axes. The first is the *spatial pattern* of the state change, whether filamentary or interface/distributed, depending on whether the conductance change is localised to a narrow conductive path or extended across the active region. The second is the *physical nature* of the state variable itself: ionic distribution, structural phase, magnetic configuration, or redox population. The two axes are orthogonal: an ionic device can also be non-filamentary, so the families considered here locate a device along both axes.

Filamentary switching. This is historically the most visible family. When a voltage is applied, a narrow conductive path forms through a previously insulating thin film, and said path can later be ruptured or reformed. In *conductive-bridge* random-access memories (CBRAM), the filament is built from metal atoms displaced electrochemically from an active electrode (like silver or copper) through a solid electrolyte [68, 69]. In *valence-change* memories, which are generally based on binary transition-metal oxides, the filament is instead associated with oxygen vacancies or partially reduced metal-cation states inside the oxide lattice [66].

These devices offer high ON/OFF ratios, fast switching, and non-volatile retention; their main limitation is stochasticity, since filament formation, positioning, and rupture vary across devices and cycles [70]. They therefore work well as binary or coarse multi-level memories but are poorly suited to fine analogue representation. Panel (a) of fig. 1.7 contrasts the CBRAM and VCM variants of these devices, while panels (b)–(e) display the other four families.

Interface (non-filamentary) switching. In these devices, conductance changes over a distributed region rather than through a single localised filament. Some implementations modulate the height or width of a Schottky barrier by accumulating or depleting mobile ions near a metal–

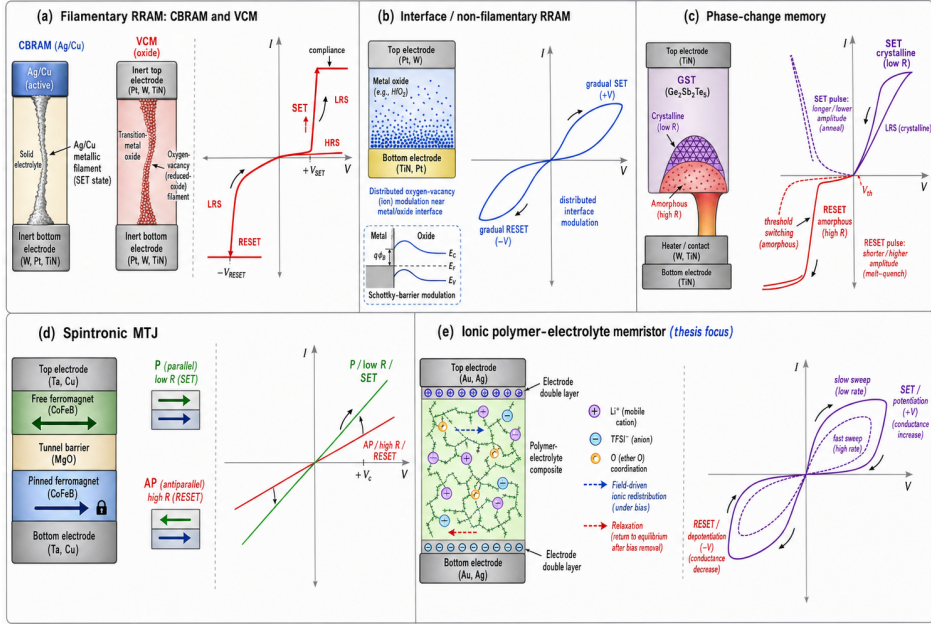


Figure 1.7: Mechanism gallery for the five memristive device families. (a) Filamentary devices, in their conductive-bridge (active-electrode metal filament) and valence-change (oxygen-vacancy filament) variants, switch abruptly between a high-resistance and a compliance-limited low-resistance branch. (b) Interface or non-filamentary devices switch through a spatially distributed change in the active layer, producing smooth, gradual, area-scaling hysteresis without an abrupt set or reset. (c) Phase-change devices toggle between amorphous and crystalline states of a chalcogenide alloy through Joule-heating melt-quench and anneal pulses, with a sharp threshold-switching event preceding the crystalline branch. (d) Spintronic magnetic tunnel junctions switch the relative orientation of two ferromagnetic layers, producing two nearly-linear branches separated by a small (order of two) ON/OFF ratio. (e) Ionic devices, in their polymer-electrolyte composite form most relevant to this thesis, redistribute mobile ions across the active layer under bias and relax back toward equilibrium when the bias is removed, producing continuous, sweep-rate-dependent hysteresis without abrupt transitions.

semiconductor interface [71]. Other implementations, especially in mixed ionic–electronic systems, drive a distributed electrochemical doping of the semiconductor [19, 20]. Given that the state change is spatially extended, these devices can support a quasi-continuous range of conductance values. The ON/OFF ratio is often smaller than that of filamentary devices, but variability is usually lower and the current tends to scale more predictably with device area. This one is the closest *spatial-pattern* class for several organic electrochemical devices, including the polymer-composite devices studied here, whose state variable is itself described under the ionic family below.

Phase-change memories. The active material is a chalcogenide alloy, usually $\text{Ge}_2\text{Sb}_2\text{Te}_5$ or a related composition, whose resistance depends on whether the material is in a crystalline or amorphous configuration. Joule-heating pulses are then used either to melt and quench the material into the amorphous state, or to anneal it back toward the crystalline state [17, 18, 72]. Phase-change memories offer excellent retention, large ON/OFF ratios, and high endurance. The cost lies in relatively large write energies and, unless the device is carefully engineered, a tendency toward binary or few-state operation. They are nonetheless mature storage-class memories and among the most important platforms for analogue in-memory computing demonstrations.

Spintronic (magnetic) devices. Their resistance depends on the magnetic configuration of a tunnel junction, typically determined by the relative orientation of two ferromagnetic layers [73]. Spintronic devices switch very quickly at low write energy and form the basis of spin-transfer-torque magnetic random-access memory. Their limitations for analogue synaptic use are a small ON/OFF ratio (often of order two) and few stable intermediate states. Even so, their intrinsic stochastic dynamics have been put to use as a computational resource within appropriately designed architectures [74, 75].

Ionic memristors. This family is the one most relevant to this thesis. Their internal state is the distribution of one or more mobile ionic species inside a solid matrix. The host can be an inorganic oxide, a chalcogenide, a halide perovskite, a mixed ionic–electronic conductor, or a polymer electrolyte. Likewise, the mobile species can be a small cation such as Li^+ , Na^+ , or Ag^+ , a proton, or a small anion; in halide perovskites it is typically a migrating halide vacancy, which has been used to build energy-efficient memristive and synaptic devices [76]. What sets this family apart is the fact that the state variable is itself a chemical distribution of mobile species, and not only a structural, morphological, or magnetic configuration of the host [69]. Its kinetics are then governed by ionic transport, coordination, and relaxation within the matrix.

Ionic systems are intrinsically history-dependent: the distribution of mobile species at any moment reflects the electric fields previously applied. They also relax once the field is removed, as intermolecular forces and concentration gradients drive the biased distribution back toward equilibrium, so volatility is intrinsic to their operation. The associated timescales are set by ion transport and binding, both tunable chemically through the host matrix and the mobile species—the link between chemistry and relaxation that leads this thesis to the organic polymer-electrolyte subfamily.

1.3.4 From Device Metrics to Fading Memory

Memristive devices are often compared through a compact set of practical figures of merit: high- and low-resistance states, ON/OFF ratio, set/reset voltage, write energy, switching speed, retention, endurance, update linearity and symmetry, and device-to-device or cycle-to-cycle variability [13, 67, 70]. These quantities are useful only after the computational role of the device has been fixed.

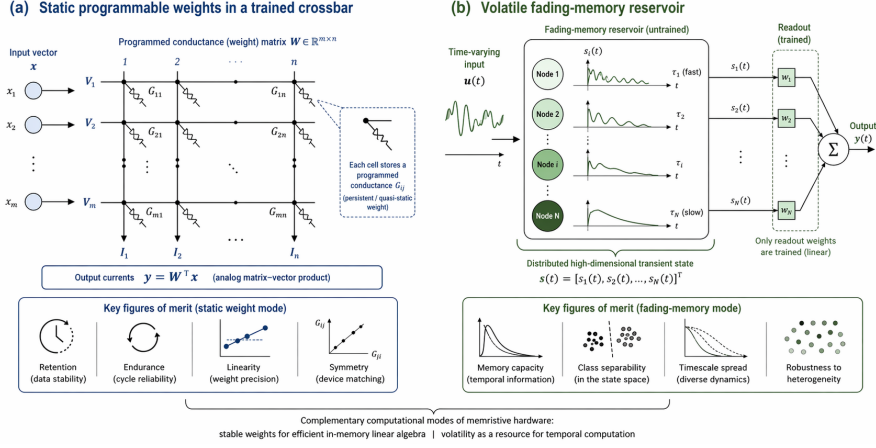


Figure 1.8: Two complementary computational roles for memristive devices. (a) A trained crossbar holds programmed conductances as non-volatile weights and performs matrix–vector multiplication by Ohm’s and Kirchhoff’s laws; here retention, endurance, linearity and symmetry are the relevant figures of merit. (b) A reservoir of volatile fading-memory nodes, driven by a time-varying input, projects temporal signals into a high-dimensional dynamical state; only a linear readout layer is trained, and the relevant figures of merit become memory capacity, class separability, and the spread of available time constants.

For analogue in-memory computing, the device stores a *trained weight*: a conductance programmed into a crossbar cell and read repeatedly during matrix–vector multiplication [12, 13]. Long retention, high endurance, low variability, and linear symmetric updates are therefore valuable because they preserve numerical accuracy. Judged by that standard, an organic or polymer-composite device with finite retention can look weak.

For temporal computing, the same finite retention becomes the operating principle. A fading-memory device should forget on a timescale matched to the input signal [15, 27]; nonlinearity can enrich the reservoir state, and device-to-device differences can supply a spread of temporal kernels. Figure 1.8 sketches the two modes schematically, and table 1.2 sets the corresponding parameter targets side by side.

Table 1.2: Direction in which each device parameter should be optimised under the trained-weight regime (section 1.3.4) and under the fading-memory regime (section 1.3.5). Several optima are not merely different in magnitude but opposed in direction, which is why a single device cannot in general satisfy both regimes simultaneously.

Parameter	Trained-weight optimum	Fading-memory optimum
Retention	As long as possible (s-years)	Matched to the input-signal timescale
Switching speed	As fast as possible	Matched to the input event rate
Write energy	Minimised (programmed once, read many)	Minimised within the fast-event budget
Conductance update linearity	Linear and uniform	Less critical; nonlinearity exploited as a feature
Update symmetry ($\pm$ pulses)	Symmetric	Less critical; asymmetry can be exploited
Cycle-to-cycle variability	Minimised	Minimised (intra-device drift remains a defect)
Device-to-device heterogeneity	Minimised	Useful as reservoir diversity [16]

Variability must also be separated into two kinds. Intra-device instability, where a device cannot reproduce its own response under identical conditions, remains a defect. Inter-device heterogeneity, by contrast, is harmful in a trained-weight array but useful in a reservoir, where devices with different time constants and nonlinear transfer functions provide a richer projection of the input than identical devices would [16].

1.3.5 Reservoir Computing with Volatile Memristors

Over the past decade, volatile memristive devices have been used as physical *leaky integrators*: their state rises under stimulation, decays after the stimulus is removed, and thereby carries a temporary record of recent input. The most important computational framework for that behaviour is physical reservoir computing [16].

In a volatile-memristor reservoir, the input is encoded as a stream of voltage pulses applied to an array of devices. Their internal states evolve in response to the stimuli and relax between them, so the array provides a nonlinear mapping of recent input history into a high-dimensional feature space. A linear readout trained on that space can then tackle temporal classification, regression, or prediction tasks. Demonstrations have been reported in diffusive metal-filament devices [31, 32], dynamic oxide memristors [33], and organic electrochemical systems [34].

Coincidence detection and multi-timescale filtering are closely related uses of the same physics. Pulses that arrive within the relaxation time of a device can accumulate, giving timing sensitivity and STDP-like responses [77]; banks of devices with different relaxation constants can process one input in parallel, with fast devices capturing sharp transients and slow devices integrating longer trends [25, 78]. In this thesis these ideas are folded into the reservoir-computing framework: coincidence

appears as one effect of the fading state, and the filter-bank idea becomes a heterogeneous reservoir whose spread of time constants is supplied by composition, chemistry, and device-to-device variation.

The central metric is therefore *memory capacity*: how well the present reservoir state reconstructs past inputs at various delays, complemented by class separability on benchmark tasks and robustness to measured variability [79]. These metrics are *task-matched*. They reward devices whose physical timescales align with the temporal structure of the signal, something that HRS, LRS, ON/OFF ratio, and retention alone do not capture.

The useful subset for this thesis is therefore narrower than the full memristor literature: ionic memristive systems whose state is a chemical distribution of mobile species, whose relaxation can be tuned, and whose volatility can be used as fading memory.

1.4 Inorganic Memristors for Neuromorphic Computing

The inorganic families that dominate neuromorphic hardware are metal-oxide resistive memories, phase-change memories, and spin-transfer-torque magnetic tunnel junctions. Each is technologically important, and together they have made memristive hardware a credible post-von-Neumann substrate. Other inorganic candidates — ferroelectric tunnel junctions and FeFETs, Mott-insulator devices, and threshold-switching selectors — are narrower in analogue synaptic operation and are left outside this comparison.

1.4.1 Metal-Oxide Memristors

The broadest and most developed inorganic family is resistive switching in transition-metal-oxide thin films, in the VCM and CBRAM variants of section 1.3.3. After the TiO_2 device [65], the field expanded to HfO_2 , TaO_x , NiO , and WO_x , many already familiar to silicon processing; back-end-of-line CMOS integration—the main practical reason for their prominence—was therefore feasible from the start and enabled large crossbar demonstrations [12, 68, 69]. As non-volatile memory their figures of merit are excellent: retention of years, nanosecond switching, endurance beyond 10^9 cycles, and sub-10 nm densities [13, 67], and the same family can also be made deliberately volatile in diffusive metal-filament form, where Ag or Cu clusters dissolve after programming [31–33, 80]. Asked to behave as an analogue synapse rather than a non-volatile weight cell, however, they are limited by filament stochasticity—an appreciable device-to-device and cycle-to-cycle spread in set/reset voltage and achieved conductance [70]—and by nonlinear, asymmetric conductance updates with only a few reliably distinguishable levels, which together cap the accuracy of in-place training on metal-oxide crossbars [13].

1.4.2 Phase-Change Memory

Phase-change memory encodes its state in the proportion of crystalline and amorphous material in a chalcogenide volume rather than in a localised filament (section 1.3.3) [17]. It is a mature storage-class technology and one of the strongest inorganic platforms for analogue in-memory computing: mixed-precision in-memory arithmetic accurate enough to solve practical linear systems [72] and neural-network training with PCM as the long-term weight store [81] have both been demonstrated, making it the most serious benchmark for trained-weight analogue applications [82]. Read as a synapse, its costs are those of a thermal-structural mechanism:

reset requires local melting, so write currents and thermal loads are large; the quenched amorphous phase drifts with time and must be compensated by reading, calibration, or algorithmic correction [18]; and crystallisation kinetics can be tuned through alloy and geometry but not with the degrees of freedom of soft matter. Its natural strength is therefore high-retention, high-endurance trained weights rather than the chemically tunable fading-memory regime pursued here.

1.4.3 Spin-Transfer-Torque Magnetic Tunnel Junctions

A spin-transfer-torque device is the magnetic tunnel junction (MTJ) of section 1.3.3, here with a thin MgO barrier and switched between parallel and antiparallel magnetisation by a spin-polarised current through a shared read/write port [73]. STT-MTJs switch in nanoseconds at femtojoule energies, exceed 10^{12} cycles, and integrate in the CMOS back end [83], and their intrinsic stochasticity has been turned into a computational resource for probabilistic computing [74, 75]. For analogue synaptic use the drawbacks are intrinsic: the parallel/antiparallel resistance ratio is small (often near two) and the stable intermediate states are few, improvable only through elaborate stacks (synthetic antiferromagnets, perpendicular-anisotropy or multilayer configurations) [83, 84]; retention and write energy trade off directly through thermal stability versus switching current [83]; and the crystalline ferromagnetic stack is incompatible with simple solution processing on arbitrary substrates. They are best viewed as fast, non-volatile, few-state stochastic elements rather than analogue, multi-timescale synapses.

1.4.4 General Limitations of Inorganic Memristors

The three families differ in mechanism and maturity but share the same limitations for the specific problem of this thesis. Their design parameters are mostly geometric, stoichiometric within narrow windows, or process-related [13, 70, 78]: there is no easy way to substitute one mobile ion for another with different binding chemistry, or to change a host so as to reshape its segmental motion. Each family can be made volatile in specific implementations—diffusive filaments [31–33, 80], amorphous-phase drift [18], superparamagnetic fluctuations [74]—but sweeping a relaxation time over orders of magnitude within one platform by tuning a single chemical variable is not natural, because vacancy drift, crystallisation, and magnetic relaxation are each fixed by the host’s intrinsic energy landscape. The fabrication routes (vacuum deposition, high-temperature annealing, top-down lithography [12, 81, 82]) are ideal for dense CMOS-integrated crossbars but restrict substrate and temperature, whereas solution-processable layers could be deposited on a far wider range of substrates and reformulated directly from solution [78]. And in these systems useful inter-device diversity and harmful intra-device instability are difficult to separate [13, 18, 70], whereas a family in which heterogeneity is introduced deliberately through composition—each device staying stable under repeated operation—would give a cleaner route to the controlled variability that reservoir computing can exploit [16].

In short, analogue, ion-mediated, chemically tunable, timescale-rich behaviour on soft, solution-processable substrates is not the natural strength of the inorganic families. That gap motivates the organic ion-mediated route. Figure 1.9 summarises the qualitative comparison, while table 1.3 provides the corresponding quantitative parameters.

Table 1.3: Typical operating parameters for the inorganic memristor families. Diffusive (volatile) metal-filament devices are listed separately because they are deliberately operated in the fading-memory rather than the non-volatile regime. Numerical ranges are conservative orders of magnitude drawn from the indicated reviews; individual devices may sit outside them. “cont.” = quasi-continuous conductance.

Family	Switching	Retention	ON/OFF	Endurance	Energy/write	Analogue states
VCN [66, 67, 70]	1–100 ns	years	10^{-10^3}	10^9-10^{12}	0.1–10 pJ	4–8
CBRAM [13, 67]	1–100 ns	years	10^4-10^6	10^8-10^{10}	0.1–10 pJ	2–4
PCM [17, 18, 72, 81]	10–100 ns	years (drifts)	10^2-10^3	10^8-10^{10}	10–100 pJ	8–16
STT-MTJ [74, 75]	1–10 ns	years	2–3	$> 10^{12}$	0.1–1 pJ	2
Diffusive (volatile) [31–33]	μ s–ms	ms–s	10^{-10^3}	$\sim 10^6$	0.1–10 pJ	cont. (volatile)

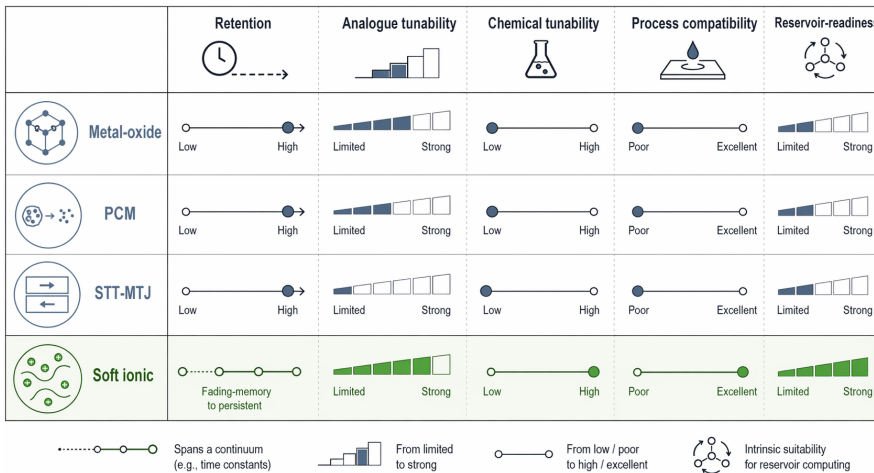


Figure 1.9: Comparison of inorganic memristor families with the soft-ionic composite route pursued in this thesis. For each family, qualitative assessments are shown along five axes: retention (the range of fading-memory to non-volatile behaviour that has been demonstrated), analogue tunability (the number of reliably programmable intermediate states), chemical tunability (the extent to which composition acts as an independent handle on the switching physics), substrate compatibility (vacuum/high-temperature workflows versus solution-processable and mild-temperature workflows), and reservoir-readiness (suitability as a fading-memory node in a temporal-computing scheme).

1.5 Organic Materials for Memristive Applications

The inorganic platforms each solve part of the problem, but none of them can be tuned by chemistry alone: reshaping the switching physics requires redesigning a rigid multilayer stack rather than changing a molecule, a host, a salt, or a mixing ratio [21, 78]. Organic and polymeric materials lift that constraint. A semiconducting polymer can be tuned through backbone, side chains, and molecular weight [85]; an ion-transport host through its chelating groups, topology, and glass-transition temperature [86, 87]; and a mobile ion through charge, radius, and Lewis acidity [88]. Most such

active layers are also solution-processable and processed below 150 °C, so a new formulation can be prepared by changing component concentrations in solution, and the mild processing is compatible with flexible, conformal, and biological substrates—as the stretchable artificial afferent nerve of Kim et al. [89] and the *nervetronic* systems reviewed by Park and co-workers [90] illustrate.

Two consequences matter most here. First, organic switching can be distributed over many molecular sites rather than confined to a single filament, which makes graded analogue states more accessible [19, 20]. Second, relaxation timescales can be engineered *chemically*: ion transport in a polymer electrolyte depends on how strongly the mobile ion binds the host, on host segmental mobility, and on the number of mobile species [86, 91], so timescales from milliseconds to many seconds—the range of biological short-term plasticity and slow temporal signals [41, 42]—can be reached within one material family by compositional change. Cost, reproducibility, biocompatibility, and lifetime still have to be shown case by case; the useful target is therefore not "organic" in general, but a reversible mixed ionic–electronic platform whose chemistry controls the fading-memory state.

1.5.1 Organic Mechanisms as Design Choices

As with their inorganic counterparts, the term *organic memristor* names a material family rather than a single mechanism. The relevant comparison is what each mechanism — each with its own state variable, reversibility, timescale, and degree of analogue control — offers for a chemically tunable, ion-mediated, two-terminal synaptic element.

Charge trapping and detrapping. Deep or shallow traps—molecular defects, embedded nanoparticles, or boundaries between incompatible polymer phases—are filled and emptied under bias, and the trap occupa-

tion, through its local space-charge environment, sets the conductance. Trapping is common in disordered organic semiconductors and operates at modest voltages, but its analogue control is limited: the state is often binary [92], the conductance can depend non-monotonically on occupation, the trap identity is hard to establish, and the retention (set by trap depth relative to thermal energy) is not easily tuned by composition. Panel (a) of fig. 1.10 sketches this trap-occupation mechanism.

Redox switching. A redox-active species—a discrete molecule in a polymer host, a redox polymer, or a unit on a conjugated backbone—is oxidised or reduced under bias, and the distribution of oxidation states sets the conductance. The state variable is then a chemical oxidation state with experimentally accessible potentials, reversible over many cycles in favourable cases; important examples include redox-active conjugated-polymer systems [93–95] and early electrochemical devices based on coordination complexes [96]. Not every redox process is cleanly reversible, though—side reactions, dimerisation, and slow degradation limit endurance—and purely redox devices tend to be bistable unless the response is distributed over many sites.

Conformational (structural) switching. In this case, the active layer contains a molecular species whose geometry, isomerisation state, or aggregation state can be modulated by an electric field, local heating, or coupled electrochemical processes. Examples include photochromic and electrochromic switches, as well as bistable molecular architectures whose conductance depends on molecular conformation [94, 95]. The state of these systems is related to a well-defined structural change, so they offer a high level of molecular design precision. However, the synthesis can be challenging, and the switching timescale is fixed by a particular molecular rearrangement.

Ionic redistribution in mixed ionic–electronic systems. This is the family most relevant to this thesis. The active layer combines an electronic

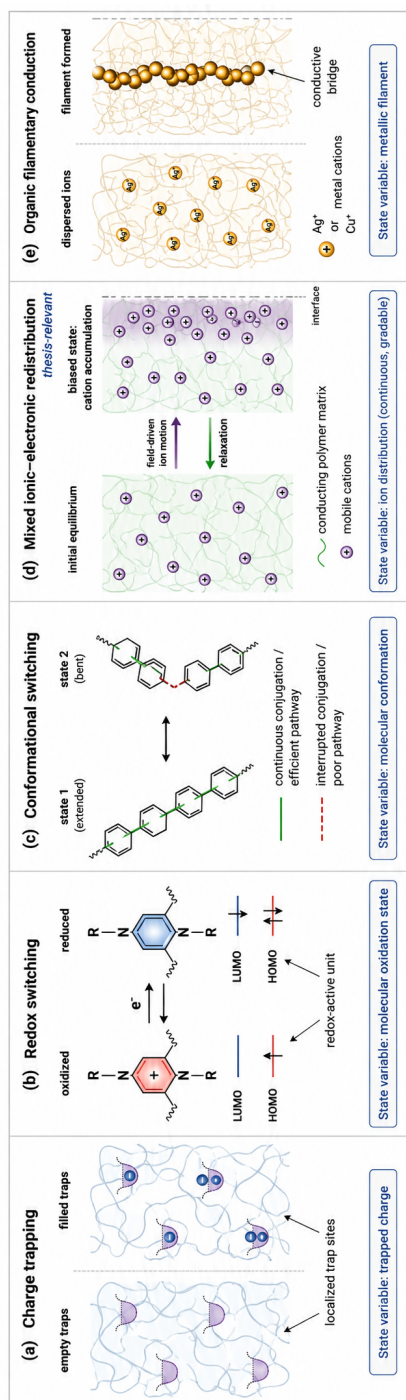


Figure 1.10: Molecular-level representation of the five mechanism families behind organic memristive behaviour. (a) Charge trapping: localised trap sites associated with defects, embedded nanoparticles, or interphase boundaries in a disordered polymer matrix, filled and emptied under bias. (b) Redox switching: a redox-active moiety alternating between its oxidised and reduced states, with the local electronic structure changing accordingly. (c) Conformational switching: a molecular unit toggling between two well-defined geometric or isomeric states with different conjugation or transport pathways. (d) Mixed ionic-electronic redistribution: a population of mobile cations within an electronically conducting polymer matrix, displaced and accumulated near an interface under bias and relaxing back toward equilibrium when the bias is removed. (e) Organic filamentary conduction: metallic cations (typically Ag^+ or Cu^+) from an active electrode migrating through the organic matrix to form a conductive bridge.

conductor—usually a conjugated polymer—with mobile ions held in an ion-conducting host. Under bias the ions move and accumulate near interfaces, modulating electronic transport through interfacial doping, electrostatic band bending, or electrochemical doping; the conductance change is gradual, multi-state, and directly tied to the amount and arrangement of displaced ionic charge [21]. The same physics underlies light-emitting electrochemical cells (LECs) [97–103], where redistribution forms a doped junction, and organic electrochemical transistors (OECTs) [104], where a gate drives the ion exchange. In the two-terminal memristive mode the driving bias itself plays the role of the gate, and the residual ionic distribution left after the bias is removed is what carries the memory of the stimulus.

The state variable is thus the distribution of mobile ions in the host, and relaxation after bias—governed by ion transport, the polymer binding environment, salt concentration, and morphology—sets the fading-memory process. The practical costs are that ionic motion is slower than electronic switching and that side reactions with the electrodes must be suppressed through chemistry and contact choice [105].

Organic filamentary conduction. Metallic species (typically Ag^+ or Cu^+ from an active electrode) migrate through the organic matrix and form conductive bridges similar to those found in inorganic CBRAM [68, 69]. These devices inherit the strengths of filamentary switching, such as large ON/OFF ratios and low switching voltages, but also its weaknesses, like stochastic filament formation and intra-device instability [106, 107].

The five families and their trade-offs are collected in table 1.4.

Table 1.4: Mechanism families behind organic memristive behaviour.

Family	State variable	Reversibility	Analogue gradation	Timescale tunability	Refs.
Charge trapping	Trap occupation	Partial	Limited (often binary)	Low	[92]
Redox	Oxidation state of a moiety	Variable	Bistable to few-state	Low–medium	[93, 96]
Conformational	Molecular geometry / isomer	Good (per design)	Bistable / few-state	Low (per molecule)	[94, 95]
Mixed ionic–electronic	Mobile-ion distribution	Good	Continuous	High (host chemistry)	[19–21]
Filamentary (organic CBRAM)	Metallic filament	Limited (stochastic)	Few states	Low	[106, 107]

1.5.2 Organic Precedents for the Composite Strategy

The relevant precedents are not a chronological checklist but a set of partial solutions to the same materials problem: how to combine ion-mediated conductance control, analogue reversibility, and a simple device geometry. The earliest two-terminal organic memristive devices are usually taken to be the polyaniline (PANI) electrochemical junctions of Erokhin, Berzina, and co-workers [108–111], in which a PANI channel interfaced with a poly(ethylene oxide)-based electrolyte is doped electrochemically by ionic redistribution under bias. This work already paired a conjugated polymer with an ion-conducting polyether host—essentially the materials logic extended throughout this thesis—and framed the device from the start as an adaptive, analogue-tunable element supporting history-dependent conductance changes.

The LEC route is especially close to the platform used here. Zakhidov’s *light-emitting memristor* [96], in which an ionic $[\text{Ru}(\text{bpy})_3]^{2+}$ complex supplied both the emitter and the mobile ions, showed that redistribution in a polymer host yields memristive behaviour, while resistive switching in simple films such as poly(vinyl alcohol) [92] confirmed that the phenomenon is accessible in cheap matrices but typically binary and mechanistically opaque. The best analogue control came from the electrochemical-transistor route: Gkoupidenis’ PEDOT:PSS OECTs reproduced short-term plasticity, paired-pulse facilitation, and long-term potentiation and depression [104, 112], Xu’s core–sheath nanowire transistors [113] and the Van de Burgt–Fuller organic electrochemical neuromorphic device (ENODE) [19, 20] delivered linear, symmetric updates through ion exchange between electrolyte and channel [21], and halide-perovskite-gated OECTs extended the accessible timescales [114]. Each such synaptic element, however, requires a third terminal and a contacted

electrolyte volume, which makes a self-contained two-terminal device an attractive complement for the reservoir paradigms of section 1.3.5.

The two-terminal line developed in parallel. Liu’s biomimicking redox organic memristor [93] demonstrated coexisting short- and long-term memory in a passive two-terminal device, but also exposed the redox route’s partial irreversibility, with the long-term regime sometimes involving permanent chemical change rather than controlled state evolution [115]. Molecular memristors, especially Goswami’s metal-coordinated azo-aromatic layers [94, 95], achieved contrast, endurance, and multi-state operation through a demanding synthesis; stretchable and flexible devices [89, 90, 116, 117] demonstrated mechanical compliance; and perovskite–organic hybrids [76, 118] and organic electrochemical reservoir computing [34] broadened the landscape further, the latter showing that mixed ionic–electronic devices can serve as volatile physical-reservoir elements.

By the early 2020s the field had matured into a subfield consolidating around shared physical principles [21, 119], with the individually demanding features each demonstrated—ion-mediated operation in two- and three-terminal forms, analogue symmetric programming, short-to-long-term transitions, mechanical compliance, molecular precision—but not yet combined in one simple platform. The three-terminal route bought reversibility and analogue control at the cost of architectural complexity, the two-terminal redox route simplicity at the cost of full reversibility, and the molecular route precision at the cost of synthesis. In that 2022 publication context, the specific gap addressed by the proof-of-concept chapter was a two-terminal, reversible, multi-state, ion-mediated device with interpretable chemistry built from standard organic-electronic components. The route pursued here fills that gap with a composite of a conjugated semiconducting polymer, an ion-conducting polymer host, and a dissociated alkali-metal salt. These parallel developments are collected

in fig. 1.11.

That framing should not be read as a frozen 2026 state-of-the-art claim. Since the proof-of-concept publication, organic neuromorphic electronics has advanced quickly, especially through organic electrochemical neurons and circuits for perception and biointerfacing [120], organic electrochemical-device systems for multimodal robotic learning [121], and organic-iontronic memristors for artificial synapses [122]. Those developments strengthen the larger premise that mixed ionic–electronic organic devices are useful neuromorphic substrates. They do not, however, change the role of the present thesis, which is narrower: to use a self-contained two-terminal polymer-electrolyte composite as a chemically tunable fading-memory element for temporal computing.

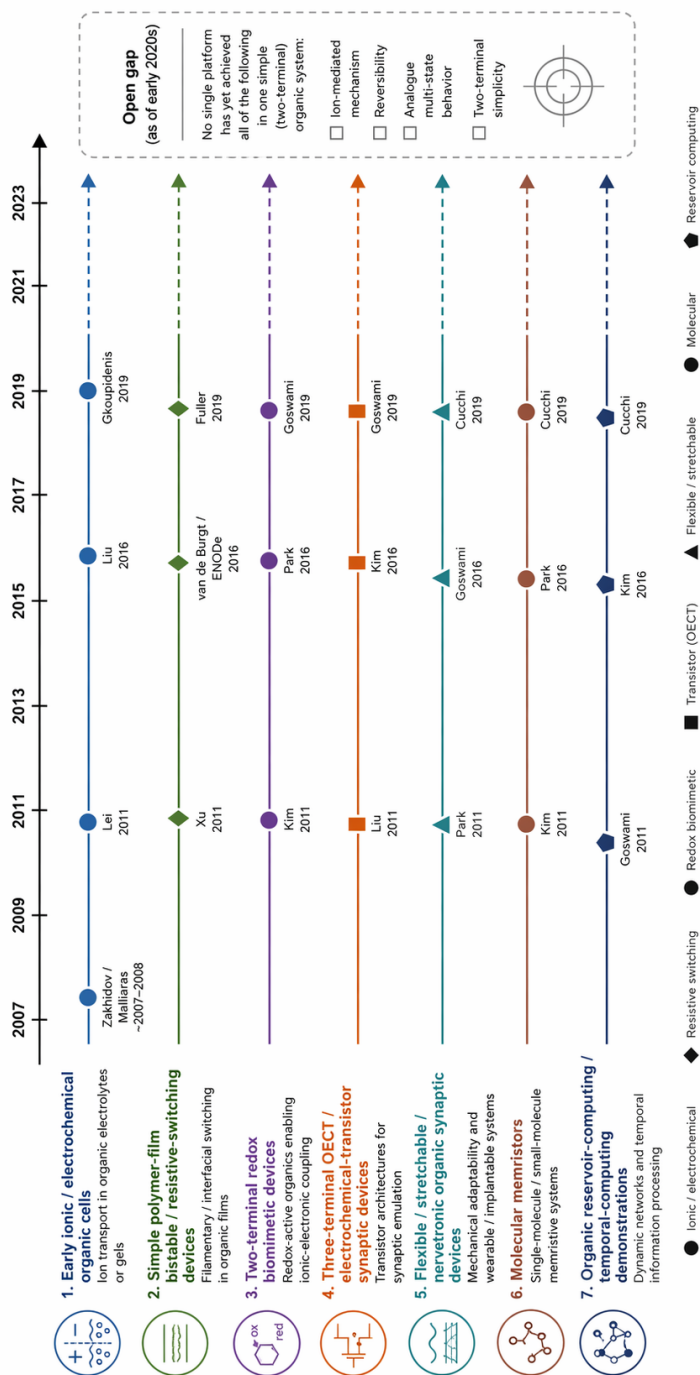


Figure 1.11: Historical map of organic memristive and organic synaptic devices, arranged by mechanism family and by year of representative publication. The map frames the literature context of the 2022 proof-of-concept publication: by the early 2020s, several demanding features of a neuromorphic organic device had each been demonstrated, but the specific combination targeted here was a self-contained two-terminal platform combining reversibility, multi-state analogue behaviour, and an interpretable ion-mediated mechanism.

1.6 Polymer Electrolyte Composites: The Enabling Strategy

The composite platform follows directly from the mixed ionic–electronic precedents above. It places a semiconducting conjugated polymer, an oxygen-rich ion-conducting polymer host, and a dissociated alkali-metal salt in one solution-processed active layer, then operates that layer in a standard two-terminal stack. Electronic transport, ion solvation, and mobile-charge supply are therefore assigned to separate components that can be changed independently.

1.6.1 Ion-Conducting Polymer Chemistry

An ion-conducting polymer is a macromolecular host in which dissolved or dispersed ions can move under an applied electric field without the help of a liquid solvent phase. Poly(ethylene oxide) (PEO) established the model system: alkali-metal salts were first shown to form solid complexes with PEO in 1973 [123], ionic conductivity was measured in those complexes in 1975 [124], and the resulting solid polymer electrolytes became practical ion conductors through the late 1970s and early 1980s [86, 87, 91]. Their relevance here is not battery performance but control of cation coordination, segmental motion, host crystallinity, and composition—the same variables that set memristive relaxation.

The polyether host has a high density of ether oxygens along the backbone or side chains. In PEO, those oxygens appear regularly along the $-\text{CH}_2-\text{CH}_2-\text{O}-$ repeat unit, and the chain conformation in the amorphous phase allows each cation to coordinate several ether oxygens from one or more polymer chains [91, 125]. Hybrane-type hyperbranched polyester-amide hosts arrange the oxygen density differently, through ether and ester groups attached as side chains on a branched polymer core [126], but the

coordination chemistry is of the same kind.

In both hosts, the oxygen atoms play the same chemical role. They act as hard Lewis bases that solvate alkali cations, stabilise salt dissociation, and define the local coordination environment of the cation. The anion, in contrast, is usually held more weakly (mainly through electrostatic compensation) and tends to be more mobile than the cation within the same host [127]. This is especially the case for soft, charge-delocalised anions such as triflate (CF_3SO_3^-) or bis(trifluoromethanesulfonyl)imide (TFSI, $\text{N}(\text{SO}_2\text{CF}_3)_2^-$).

Ion transport in these hosts is usually described through three related microscopic phenomena:

- From the *vehicular* side, a cation and its solvation shell move together through the polymer matrix, so long-range transport is coupled to polymer motion [128].
- From the *hopping* or site-to-site side, the cation instead transfers between coordination sites through a local rearrangement of oxygens, without macroscopic motion of the host. This last view is conceptually close to Grotthuss-type transport in proton conductors [129], even though the chemistry of the transfer itself is different.
- Finally, from the *segmental-motion-assisted* side, short polymer segments reorganise around the cation, creating and destroying coordination sites that allow effective diffusion [130, 131].

These three phenomena are not mutually exclusive, although the weight of each contribution during transport depends on host chemistry, cation identity, salt concentration, and temperature [86, 91].

Examining device dynamics more closely, the macroscopic ionic mobility of a polymer electrolyte generally correlates with the segmental mobility

of the host material. This segmental mobility is determined by the glass transition temperature T_g and by whether the host remains amorphous during operation. Below T_g , segmental motion is essentially frozen and ionic conduction drops. Above T_g , it is activated, and ionic transport becomes measurable, with the temperature dependence in this regime typically following the Vogel-Fulcher-Tammann form characteristic of fragile glass-formers [86, 132].

Even above T_g , however, not all of the host contributes equally. Crystalline regions contribute little to conduction and can even trap ions in ordered environments [125, 131]; the textbook picture is that conduction proceeds essentially through the amorphous fraction, although it has since been shown that specific crystalline stoichiometric phases can themselves support measurable ionic motion [133]. Low-crystallinity or fully amorphous hosts are therefore preferred for ion transport. A low T_g and a low crystalline fraction can be pursued through copolymerisation, branching, or plasticisation; hyperbranched architectures such as the Hybrane family are attractive here because branching itself suppresses crystallinity [134]. For room-temperature memristive operation the host should remain amorphous and mobile under ambient conditions, ideally without added plasticisers.

The strength of cation binding depends directly on the identity of the cation. For this matter, the concept of hard-soft acid-base (HSAB) reasoning [88] becomes important. Ether and ester oxygens in polyether and polyester-amide hosts are hard Lewis bases, meaning that their donor pairs are localised, their polarisability is low, and their coordination to alkali cations is primarily electrostatic. Among the alkali cations, Li^+ is the hardest Lewis acid (small, with a high charge density), and should therefore interact most strongly with hard oxygen donors. Na^+ and K^+ are progressively softer, with larger radii, lower charge densities, and weaker electrostatic coupling to the same oxygen environment.

This means that, in a common polyether host and at fixed composition, the residence time of the cation at a coordination site might be expected to follow the coordination-strength order $\text{Li}^+ > \text{Na}^+ > \text{K}^+$. That expectation provides a hypothesis for the devices' fading-memory timescale: harder cations should relax more slowly, softer cations more quickly. The experimental chapters test this hypothesis rather than assume it; in the archive used here, composition is the replicated quantitative variable, while cation order is not robust across host and anion. The coordination geometry, the three transport mechanisms, and the cation–oxygen binding hierarchy that motivate the hypothesis are summarised in fig. 1.12.

A second set of design parameters is provided through composition. A polymer electrolyte is defined not only by host and ion identity, but also by the salt-to-host ratio; in a composite device, the polymer-to-semiconductor ratio plays a role as well. The salt-to-host ratio sets the number and spatial density of occupied coordination sites, while the host-to-semiconductor ratio sets the volume available for ionic transport and the connectivity of the electronic phase. The two are therefore treated as independent composition axes in the later chapters.

These considerations motivate the host choices made in this thesis. Hybrane DEO750 8500 is a hyperbranched polyester-amide with a high density of ether and ester oxygens. It provides an amorphous, low-crystallinity ion-transport host and is used in ?? as the host of the proof-of-concept Super Yellow (SY)/Hybrane/lithium triflate (LiOTf , LiCF_3SO_3) composite, where it supports the full synaptic functionality demonstrated in a single device.

PEO, in turn, acts as the central host for the broader comparative work. Its chemical role is similar to that of Hybrane: ether oxygens coordinate alkali-metal cations, segmental motion enables transport, and composition can be varied systematically while the electronic semiconductor is held fixed. The reason for choosing it as the basis of the comparative

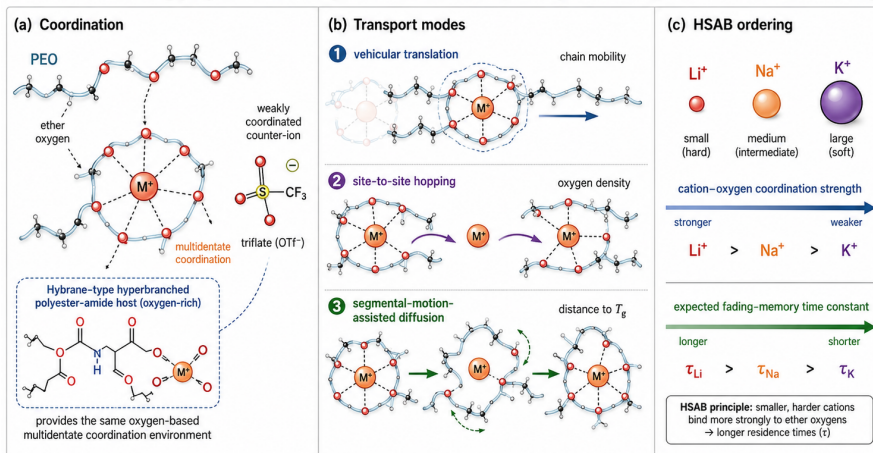


Figure 1.12: Molecular-level chemistry of ion transport in the polyether-hosted polymer-electrolyte composite. (a) A short poly(ethylene oxide) segment coordinates an alkali cation through four to six ether oxygens, with the triflate anion held nearby as a weakly coordinated charge-compensating counter-ion; the same coordination chemistry applies to the hyperbranched polyester-amide hosts used in ?? [91, 125, 127]. (b) Long-range ionic transport in this host proceeds by three coupled contributions: vehicular translation of the cation together with its solvation shell, site-to-site hopping between coordination pockets through local oxygen-cage rearrangement, and segmental-motion-assisted diffusion in which short polymer segments reorganise to create and destroy coordination sites [128, 130, 131]. (c) Within a common polyether host, hard-soft acid-base reasoning ranks the alkali cations along the $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ series in decreasing cation-oxygen coordination strength [88]. This ordering motivates the cation-timescale hypothesis tested later, but the thesis does not assume it as a host- and anion-independent law.

corpus, however, is not just chemical analogy. PEO is the most thoroughly characterised solid polymer electrolyte in the literature, with its phase diagrams, coordination geometries, conduction mechanisms, and salt-dependent dynamics having been mapped across decades of work [86, 91, 124, 125, 131]. Basing the replicated composition series on PEO therefore places the measured dynamics on a host whose structural and transport behaviour is already independently constrained. A hyperbranched

polyether, trimethylolpropane ethoxylate (TMPE), is used alongside PEO as a contrasting host in that comparison, so that the linear and branched polyether architectures can be set against each other directly. Host, anion, and cation substitutions are surveyed at smaller sample sizes rather than treated as material laws.

1.6.2 LEC-Derived Composite Operation

The coupling between ionic redistribution in a polymer host and electronic transport in a conjugated semiconductor can be both stable and reversible, and polymer light-emitting electrochemical cells (LECs) are the clearest precedent for it [135]. An LEC is a simple two-terminal sandwich [97]: a conjugated light-emitting polymer is mixed from a common solution with an ion-conducting polymer and a dissociated salt. Under bias, cations migrate toward the cathode and anions toward the anode, forming doped regions that inject carriers into the undoped bulk, and the resulting dynamic p-i-n profile produces electroluminescence in the intrinsic region between the doped edges [100].

Two models were historically used to describe this operation. In the electrochemical-doping model [97, 136], the dominant effect is bulk doping of the conjugated polymer; in the electrodynamic model [98, 99], it is the lowering of charge-injection barriers by ion accumulation at the contacts. Later experiments [100, 102] and a unifying framework [101] showed these to be limiting cases of the same ion redistribution under different injection regimes, not contradictory mechanisms. The two models, their reconciliation, and the bias-off configuration that the memristive composite reuses are compared on a common cathode–anode position axis in fig. 1.13; studies of electrochemical side reactions [105] additionally identify the degradation pathways that must be suppressed.

For the present purpose, the lessons are twofold. First, this coexistence

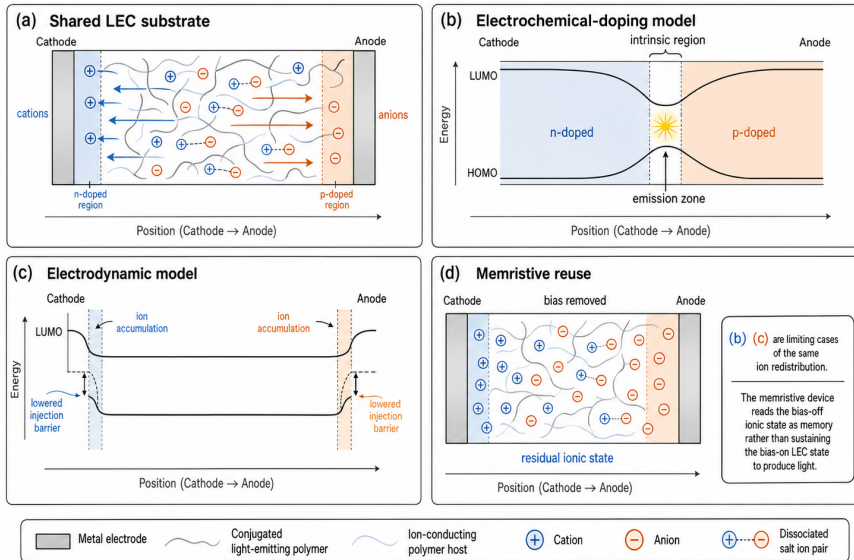


Figure 1.13: Light-emitting electrochemical cell operating models and their memristive reuse, drawn on a common cathode–anode position axis. (a) Shared substrate: under sustained bias, cations drift toward the cathode and anions toward the anode, forming doped regions at the two contacts. (b) Electrochemical-doping model: the dominant effect of the ionic redistribution is *bulk doping* of the conjugated polymer, producing a spatially wide p-i-n profile with emission localised at a narrow intrinsic region. (c) Electrodynamic model: the dominant effect of the same redistribution is *injection-barrier lowering* at the metal/polymer interfaces, producing thin ion-accumulation layers at the contacts and an essentially flat-band bulk. The two pictures were later reconciled as limiting cases of the same redistribution under different injection regimes. (d) Memristive reuse: with the bias removed, the residual ionic distribution left behind by the previous bias is the *stored internal state* of the device.

is durable: LECs operate over useful windows with lifetimes of hundreds to thousands of hours [103, 137], without destructive side reactions within a few cycles, provided the contacts are engineered—through metal choice and operating protocol—for repeated charge injection and withdrawal [138–140]. Second, the electrolyte plays the same role in both devices, supplying mobile ions whose redistribution modulates the conju-

gated polymer rather than conducting electrons itself; only its *use* differs. An LEC holds the bias on so the p-i-n profile sustains emission, whereas a memristive device writes an ionic distribution and reads the residual distribution left after the bias is removed as its stored state. That same redistribution can therefore plausibly serve as the internal state variable of a two-terminal memory device.

The most direct precedent is the Hybrane-based composite that underlies the proof-of-concept device of ???. Mardegan and co-workers [103] demonstrated stable LECs based on Hybrane hyperbranched polyester-amide hosts blended with a conjugated light-emitting polymer, reporting long operational lifetimes under continuous bias. Their work validates the semiconducting-polymer / ion-conducting-host / salt architecture, the solvent system, and the processing protocol that ??? then reuses under pulsed bias to operate the same composite as a memristive device.

1.6.3 Rationale for the Composite Approach

The composite assigns the three roles—electronic transport, ion solvation and transport, and mobile-charge supply—to separate components rather than asking a single material to perform all three [97]: a conjugated polymer, an oxygen-rich host, and a dissociated salt, so that salt, cation, or blend ratio can be changed independently of the rest of the stack [21]. The salt is the clearest control variable: a small mass fraction that fixes the identity and density of the mobile ions writing the internal state. By the HSAB argument of section 1.6.1, the harder alkali cations bind ether oxygens more strongly and should be slower to displace, whereas charge-delocalised anions such as triflate interact only weakly and redistribute more easily [127]. This mobility asymmetry is the proposed origin of the device’s memory: under bias both species move but relax at different rates, so a fast component (mobile anions, near the contacts) and a slower

component (more tightly bound cations) let a single device retain traces of past stimulation over multiple timescales [108, 110, 111], in analogy with the short- and long-term plasticity of section 1.2.2. The cation is therefore a natural hypothesis for the slow response, but not a conclusion that can be assumed independently of host, anion, electrode, and drive protocol.

Composition adds a second, more robust level of control. The salt fraction sets the number of mobile ions and hence the achievable size of the conductance change [86, 87], while the host fraction sets the volume and connectivity of the segmentally mobile phase in which they rearrange; in principle these vary the amplitude and the rate of the response separately, though the comparative chapter treats their independence as an experimental question. The architecture is also experimentally convenient, since the three components share a common solvent: a new formulation is just a change of solution ratios deposited on the same electrode geometry, with no redesign of the stack. This makes a replicated composition grid realistic, while host, anion, and cation substitutions are surveyed as lower-power chemical contrasts.

Devices with the same polymers but different blend ratios should exhibit a spread of ionic relaxation times, with sample-limited chemistry and ordinary device-to-device variation adding further diversity. In a common circuit this family acts as a bank of temporal kernels with distinct decay constants [16, 79]—a liability for storing a precisely trained weight, but a resource for covering a range of fading-memory timescales [31–34]. The application chapter develops this idea quantitatively from the measured device behaviour.

The experimental chapters test this rationale first with the Hybrane/LiOTf proof-of-concept device of ??, and then with the broader PEO-based corpus in which composition is varied as the replicated axis and chemistry is surveyed as illustrative side evidence. Figure 1.14 summarises this

concept used throughout the thesis, linking the two-terminal stack to the qualitative timescale ladder originally expected from cation and composition chemistry.

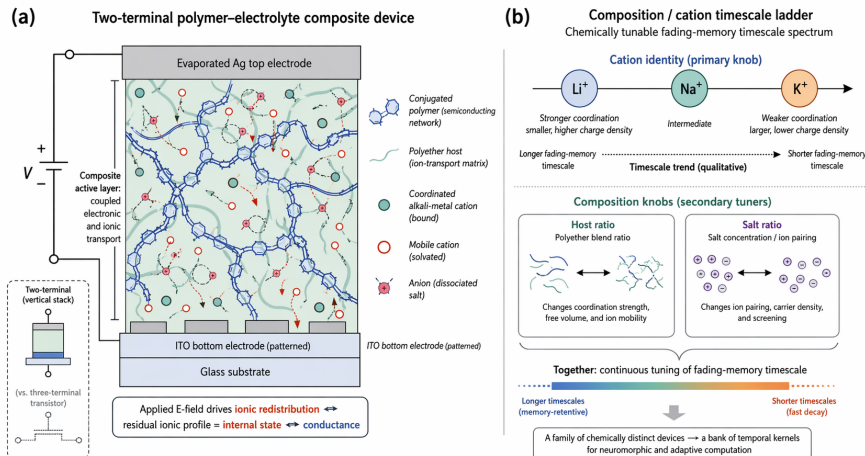


Figure 1.14: The thesis-specific polymer-electrolyte composite device and its chemically tunable dynamical timescale. (a) Vertical two-terminal stack: a solution-processed composite active layer, combining a conjugated semiconducting polymer network with an oxygen-rich ion-conducting host, coordinated alkali-metal cations, and a dissociated salt, is sandwiched between a patterned ITO bottom electrode and an evaporated silver top electrode; the internal state of the device is set by the residual ionic distribution that remains after a driving bias is removed. (b) Qualitative timescale ladder on which cation identity and the host/salt composition ratios were expected to place the fading-memory time constant of the device. ?? tests this expectation and shows that composition is the replicated quantitative variable, while the cation ordering is not robust across the available host/anion families.

From Background to Device

The resulting device target is specific: a solution-processed composite of a conjugated semiconducting polymer, an oxygen-rich ion-conducting host, and a dissociated alkali-metal salt, operated as a two-terminal device

whose residual ionic distribution is its internal state. The thesis-level motivation, scientific questions, objectives, and chapter structure are set out in the front-matter roadmap rather than repeated here. The next chapter takes the first step in that programme: the proof-of-concept SY/Hybrane/LiOTf device of ??, its memristive and synaptic functionality, and the mechanistic interpretation that paves the way for the comparative and computational work that follows.

References

1. Horowitz, M. *1.1 Computing's energy problem (and what we can do about it)* in *2014 IEEE International Solid-State Circuits Conference Digest of Technical Papers (ISSCC)* (2014), 10–14. doi:10.1109/ISSCC.2014.6757323.
2. Hennessy, J. L. & Patterson, D. A. A new golden age for computer architecture. *Communications of the ACM* **62**, 48–60. doi:10.1145/3282307 (2019).
3. LeCun, Y., Bengio, Y. & Hinton, G. Deep learning. *Nature* **521**, 436–444. doi:10.1038/nature14539 (2015).
4. Abbott, L. F. & Regehr, W. G. Synaptic computation. *Nature* **431**, 796–803. doi:10.1038/nature03010 (2004).
5. Hennessy, J. L. & Patterson, D. A. *Computer Architecture: A Quantitative Approach* 6th ed. (Morgan Kaufmann, Cambridge, MA, 2017).
6. Wulf, W. A. & McKee, S. A. Hitting the memory wall: implications of the obvious. *ACM SIGARCH Computer Architecture News* **23**, 20–24. doi:10.1145/216585.216588 (1995).
7. Von Neumann, J. First Draft of a Report on the EDVAC. *IEEE Annals of the History of Computing* **15**, 27–75. doi:10.1109/85.238389 (1993).

REFERENCES

8. Editorial. Big data needs a hardware revolution. *Nature* **554**, 145–146. doi:10.1038/d41586-018-01683-1 (2018).
9. Moore, G. E. Cramming more components onto integrated circuits. *Electronics* **38**, 114–117 (1965).
10. Dennard, R. H., Gaensslen, F. H., Yu, H.-N., Rideout, V. L., Bassous, E. & LeBlanc, A. R. Design of ion-implanted MOSFET's with very small physical dimensions. *IEEE Journal of Solid-State Circuits* **SC-9**, 256–268. doi:10.1109/JSSC.1974.1050511 (1974).
11. Esmailzadeh, H., Blem, E., St. Amant, R., Sankaralingam, K. & Burger, D. *Dark silicon and the end of multicore scaling in Proceedings of the 38th Annual International Symposium on Computer Architecture (ISCA '11)* (2011), 365–376. doi:10.1145/2000064.2000108.
12. Prezioso, M., Merrih-Bayat, F., Hoskins, B., Adam, G. C., Likharev, K. K. & Strukov, D. B. Training and operation of an integrated neuromorphic network based on metal-oxide memristors. *Nature* **521**, 61–64. doi:10.1038/nature14441 (2015).
13. Ielmini, D. & Wong, H.-S. P. In-memory computing with resistive switching devices. *Nature Electronics* **1**, 333–343. doi:10.1038/s41928-018-0092-2 (2018).
14. Jaeger, H. *The “echo state” approach to analysing and training recurrent neural networks – with an erratum note* tech. rep. GMD Report 148 (German National Research Center for Information Technology (GMD), Bonn, Germany, 2001).
15. Maass, W., Natschläger, T. & Markram, H. Real-Time Computing Without Stable States: A New Framework for Neural Computation Based on Perturbations. *Neural Computation* **14**, 2531–2560. doi:10.1162/089976602760407955 (2002).

-
16. Tanaka, G., Yamane, T., Héroux, J. B., *et al.* Recent advances in physical reservoir computing: A review. *Neural Networks* **115**, 100–123. doi:10.1016/j.neunet.2019.03.005 (2019).
 17. Koelmans, W. W., Sebastian, A., Jonnalagadda, V. P., Krebs, D., Dellmann, L. & Eleftheriou, E. Projected phase-change memory devices. *Nature Communications* **6**, 8181. doi:10.1038/ncomms9181 (2015).
 18. Sebastian, A., Le Gallo, M. & Krebs, D. Crystal growth within a phase change memory cell. *Nature Communications* **5**, 4314. doi:10.1038/ncomms5314 (2014).
 19. Fuller, E. J., Keene, S. T., Melianas, A., *et al.* Parallel programming of an ionic floating-gate memory array for scalable neuromorphic computing. *Science* **364**, 570–574. doi:10.1126/science.aaw5581 (2019).
 20. Van de Burgt, Y., Lubberman, E., Fuller, E. J., *et al.* A non-volatile organic electrochemical device as a low-voltage artificial synapse for neuromorphic computing. *Nature Materials* **16**, 414–418. doi:10.1038/nmat4856 (2017).
 21. Van de Burgt, Y., Melianas, A., Keene, S. T., Malliaras, G. & Salleo, A. Organic electronics for neuromorphic computing. *Nature Electronics* **1**, 386–397. doi:10.1038/s41928-018-0103-3 (2018).
 22. Yao, P., Wu, H., Gao, B., *et al.* Fully hardware-implemented memristor convolutional neural network. *Nature* **577**, 641–646. doi:10.1038/s41586-020-1942-4 (2020).
 23. Vaswani, A., Shazeer, N., Parmar, N., *et al.* *Attention Is All You Need* in *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)* (2017), 5998–6008. doi:10.48550/arXiv.1706.03762.

REFERENCES

24. Roy, K., Jaiswal, A. & Panda, P. Towards spike-based machine intelligence with neuromorphic computing. *Nature* **575**, 607–617. doi:10.1038/s41586-019-1677-2 (2019).
25. Indiveri, G. & Liu, S.-C. Memory and Information Processing in Neuromorphic Systems. *Proceedings of the IEEE* **103**, 1379–1397. doi:10.1109/JPROC.2015.2444094 (2015).
26. Mead, C. Neuromorphic electronic systems. *Proceedings of the IEEE* **78**, 1629–1636. doi:10.1109/5.58356 (1990).
27. Boyd, S. & Chua, L. O. Fading memory and the problem of approximating nonlinear operators with Volterra series. *IEEE Transactions on Circuits and Systems* **32**, 1150–1161. doi:10.1109/TCS.1985.1085649 (1985).
28. Jaeger, H. & Haas, H. Harnessing Nonlinearity: Predicting Chaotic Systems and Saving Energy in Wireless Communication. *Science* **304**, 78–80. doi:10.1126/science.1091277 (2004).
29. Lukoševičius, M. & Jaeger, H. Reservoir computing approaches to recurrent neural network training. *Computer Science Review* **3**, 127–149. doi:10.1016/j.cosrev.2009.03.005 (2009).
30. Verstraeten, D., Schrauwen, B., D’Haene, M. & Stroobandt, D. An experimental unification of reservoir computing methods. *Neural Networks* **20**, 391–403. doi:10.1016/j.neunet.2007.04.003 (2007).
31. Du, C., Cai, F., Zidan, M. A., Ma, W., Lee, S. H. & Lu, W. D. Reservoir computing using dynamic memristors for temporal information processing. *Nature Communications* **8**, 2204. doi:10.1038/s41467-017-02337-y (2017).
32. Midya, R., Wang, Z., Asapu, S., *et al.* Reservoir Computing Using Diffusive Memristors. *Advanced Intelligent Systems* **1**, 1900084. doi:10.1002/aisy.201900084 (2019).

-
33. Zhong, Y., Tang, J., Li, X., Gao, B., Qian, H. & Wu, H. Dynamic memristor-based reservoir computing for high-efficiency temporal signal processing. *Nature Communications* **12**, 408. doi:10.1038/s41467-020-20692-1 (2021).
 34. Cucchi, M., Gruener, C., Petrauskas, L., *et al.* Reservoir computing with biocompatible organic electrochemical networks for brain-inspired bio-signal classification. *Science Advances* **7**, eabh0693. doi:10.1126/sciadv.abh0693 (2021).
 35. Drubach, D. *The Brain Explained* (Prentice Hall, Upper Saddle River, NJ, 2000).
 36. *Principles of Neural Science* 5th ed. (eds Kandel, E. R., Schwartz, J. H., Jessell, T. M., Siegelbaum, S. A. & Hudspeth, A. J.) (McGraw-Hill, New York, 2013).
 37. Hodgkin, A. L. & Huxley, A. F. A quantitative description of membrane current and its application to conduction and excitation in nerve. *The Journal of Physiology* **117**, 500–544. doi:10.1113/jphysiol.1952.sp004764 (1952).
 38. Rieke, F., Warland, D., de Ruyter van Steveninck, R. & Bialek, W. *Spikes: Exploring the Neural Code* (MIT Press, Cambridge, MA, 1997).
 39. Südhof, T. C. The synaptic vesicle cycle. *Annual Review of Neuroscience* **27**, 509–547. doi:10.1146/annurev.neuro.26.041002.131412 (2004).
 40. Hebb, D. O. *The Organization of Behavior: A Neuropsychological Theory* (Wiley, New York, 1949).
 41. Zucker, R. S. Short-term synaptic plasticity. *Annual Review of Neuroscience* **12**, 13–31. doi:10.1146/annurev.ne.12.030189.000305 (1989).

REFERENCES

42. Zucker, R. S. & Regehr, W. G. Short-term synaptic plasticity. *Annual Review of Physiology* **64**, 355–405. doi:10.1146/annurev.physiol.64.092501.114547 (2002).
43. Bliss, T. V. P. & Lømo, T. Long-lasting potentiation of synaptic transmission in the dentate area of the anaesthetized rabbit following stimulation of the perforant path. *Journal of Physiology* **232**, 331–356. doi:10.1113/jphysiol.1973.sp010273 (1973).
44. Bliss, T. V. P. & Collingridge, G. L. A synaptic model of memory: long-term potentiation in the hippocampus. *Nature* **361**, 31–39. doi:10.1038/361031a0 (1993).
45. Abraham, W. C. How long will long-term potentiation last? *Philosophical Transactions of the Royal Society B: Biological Sciences* **358**, 735–744. doi:10.1098/rstb.2002.1222 (2003).
46. Mayer, M. L., Westbrook, G. L. & Guthrie, P. B. Voltage-dependent block by Mg^{2+} of NMDA responses in spinal cord neurones. *Nature* **309**, 261–263. doi:10.1038/309261a0 (1984).
47. Nowak, L., Bregestovski, P., Ascher, P., Herbet, A. & Prochiantz, A. Magnesium gates glutamate-activated channels in mouse central neurones. *Nature* **307**, 462–465. doi:10.1038/307462a0 (1984).
48. Malinow, R. & Malenka, R. C. AMPA receptor trafficking and synaptic plasticity. *Annual Review of Neuroscience* **25**, 103–126. doi:10.1146/annurev.neuro.25.112701.142758 (2002).
49. Lüscher, C. & Malenka, R. C. NMDA receptor-dependent long-term potentiation and long-term depression (LTP/LTD). *Cold Spring Harbor Perspectives in Biology* **4**, a005710. doi:10.1101/cshperspect.a005710 (2012).

-
50. Dudek, S. M. & Bear, M. F. Homosynaptic long-term depression in area CA1 of hippocampus and effects of N-methyl-D-aspartate receptor blockade. *Proceedings of the National Academy of Sciences* **89**, 4363–4367. doi:10.1073/pnas.89.10.4363 (1992).
 51. Mulkey, R. M. & Malenka, R. C. Mechanisms underlying induction of homosynaptic long-term depression in area CA1 of the hippocampus. *Neuron* **9**, 967–975. doi:10.1016/0896-6273(92)90248-C (1992).
 52. Malenka, R. C. & Bear, M. F. LTP and LTD: an embarrassment of riches. *Neuron* **44**, 5–21. doi:10.1016/j.neuron.2004.09.012 (2004).
 53. Markram, H., Lübke, J., Frotscher, M. & Sakmann, B. Regulation of synaptic efficacy by coincidence of postsynaptic APs and EPSPs. *Science* **275**, 213–215. doi:10.1126/science.275.5297.213 (1997).
 54. Bi, G.-q. & Poo, M.-m. Synaptic Modifications in Cultured Hippocampal Neurons: Dependence on Spike Timing, Synaptic Strength, and Postsynaptic Cell Type. *Journal of Neuroscience* **18**, 10464–10472. doi:10.1523/JNEUROSCI.18-24-10464.1998 (1998).
 55. Herculano-Houzel, S. The Human Brain in Numbers: A Linearly Scaled-up Primate Brain. *Frontiers in Human Neuroscience* **3**, 31. doi:10.3389/neuro.09.031.2009 (2009).
 56. Attwell, D. & Laughlin, S. B. An Energy Budget for Signaling in the Grey Matter of the Brain. *Journal of Cerebral Blood Flow & Metabolism* **21**, 1133–1145. doi:10.1097/00004647-200110000-00001 (2001).

REFERENCES

57. Kuzum, D., Yu, S. & Wong, H.-S. P. Synaptic electronics: materials, devices and applications. *Nanotechnology* **24**, 382001. doi:10.1088/0957-4484/24/38/382001 (2013).
58. Chua, L. O. Memristor – The Missing Circuit Element. *IEEE Transactions on Circuit Theory* **18**, 507–519. doi:10.1109/TCT.1971.1083337 (1971).
59. Adhikari, S. P., Sah, M. P., Kim, H. & Chua, L. O. Three Fingerprints of Memristor. *IEEE Transactions on Circuits and Systems I: Regular Papers* **60**, 3008–3021. doi:10.1109/TCSI.2013.2256171 (2013).
60. Chua, L. O. If it's pinched it's a memristor. *Semiconductor Science and Technology* **29**, 104001. doi:10.1088/0268-1242/29/10/104001 (2014).
61. Chua, L. O. & Kang, S. M. Memristive devices and systems. *Proceedings of the IEEE* **64**, 209–223. doi:10.1109/PROC.1976.10092 (1976).
62. Hickmott, T. W. Low-frequency negative resistance in thin anodic oxide films. *Journal of Applied Physics* **33**, 2669–2682. doi:10.1063/1.1702530 (1962).
63. Argall, F. Switching phenomena in titanium oxide thin films. *Solid-State Electronics* **11**, 535–541. doi:10.1016/0038-1101(68)90092-0 (1968).
64. Dearnaley, G., Stoneham, A. M. & Morgan, D. V. Electrical phenomena in amorphous oxide films. *Reports on Progress in Physics* **33**, 1129–1191. doi:10.1088/0034-4885/33/3/306 (1970).
65. Strukov, D. B., Snider, G. S., Stewart, D. R. & Williams, R. S. The missing memristor found. *Nature* **453**, 80–83. doi:10.1038/nature06932 (2008).

-
66. Yang, J. J., Pickett, M. D., Li, X., Ohlberg, D. A. A., Stewart, D. R. & Williams, R. S. Memristive switching mechanism for metal/oxide/metal nanodevices. *Nature Nanotechnology* **3**, 429–433. doi:10.1038/nnano.2008.160 (2008).
 67. Yang, J. J., Strukov, D. B. & Stewart, D. R. Memristive devices for computing. *Nature Nanotechnology* **8**, 13–24. doi:10.1038/nnano.2012.240 (2013).
 68. Valov, I., Waser, R., Jameson, J. R. & Kozicki, M. N. Electrochemical metallization memories—fundamentals, applications, prospects. *Nanotechnology* **22**, 254003. doi:10.1088/0957-4484/22/25/254003 (2011).
 69. Waser, R. & Aono, M. Nanoionics-based resistive switching memories. *Nature Materials* **6**, 833–840. doi:10.1038/nmat2023 (2007).
 70. Sun, W., Gao, B., Chi, M., *et al.* Understanding memristive switching via in situ characterization and device modeling. *Nature Communications* **10**, 3453. doi:10.1038/s41467-019-11411-6 (2019).
 71. Sawa, A. Resistive switching in transition metal oxides. *Materials Today* **11**, 28–36. doi:10.1016/S1369-7021(08)70119-6 (2008).
 72. Le Gallo, M., Sebastian, A., Mathis, R., *et al.* Mixed-precision in-memory computing. *Nature Electronics* **1**, 246–253. doi:10.1038/s41928-018-0054-8 (2018).
 73. Chappert, C., Fert, A. & Van Dau, F. N. The emergence of spin electronics in data storage. *Nature Materials* **6**, 813–823. doi:10.1038/nmat2024 (2007).

REFERENCES

74. Mizrahi, A., Hirtzlin, T., Fukushima, A., *et al.* Neural-like computing with populations of superparamagnetic basis functions. *Nature Communications* **9**, 1533. doi:10.1038/s41467-018-03963-w (2018).
75. Torrejon, J., Riou, M., Araujo, F. A., *et al.* Neuromorphic computing with nanoscale spintronic oscillators. *Nature* **547**, 428–431. doi:10.1038/nature23011 (2017).
76. Xiao, Z. & Huang, J. Energy-Efficient Hybrid Perovskite Memristors and Synaptic Devices. *Advanced Electronic Materials* **2**, 1600100. doi:10.1002/aelm.201600100 (2016).
77. Bichler, O., Querlioz, D., Thorpe, S. J., Bourgoin, J.-P. & Gamrat, C. Extraction of temporally correlated features from dynamic vision sensors with spike-timing-dependent plasticity. *Neural Networks* **32**, 339–348. doi:10.1016/j.neunet.2012.02.022 (2012).
78. Sangwan, V. K. & Hersam, M. C. Neuromorphic nanoelectronic materials. *Nature Nanotechnology* **15**, 517–528. doi:10.1038/s41565-020-0647-z (2020).
79. Dambre, J., Verstraeten, D., Schrauwen, B. & Massar, S. Information Processing Capacity of Dynamical Systems. *Scientific Reports* **2**, 514. doi:10.1038/srep00514 (2012).
80. Ohno, T., Hasegawa, T., Tsuruoka, T., Terabe, K., Gimzewski, J. K. & Aono, M. Short-term plasticity and long-term potentiation mimicked in single inorganic synapses. *Nature Materials* **10**, 591–595. doi:10.1038/nmat3054 (2011).
81. Ambrogio, S., Narayanan, P., Tsai, H., *et al.* Equivalent-accuracy accelerated neural-network training using analogue memory. *Nature* **558**, 60–67. doi:10.1038/s41586-018-0180-5 (2018).

-
82. Wong, H.-S. P., Raoux, S., Kim, S., *et al.* Phase Change Memory. *Proceedings of the IEEE* **98**, 2201–2227. doi:10.1109/JPROC.2010.2070050 (2010).
83. Bhatti, S., Sbiaa, R., Hirohata, A., Ohno, H., Fukami, S. & Piramanayagam, S. N. Spintronics based random access memory: a review. *Materials Today* **20**, 530–548. doi:10.1016/j.mattod.2017.07.007 (2017).
84. Ikeda, S., Miura, K., Yamamoto, H., *et al.* A perpendicular-anisotropy CoFeB–MgO magnetic tunnel junction. *Nature Materials* **9**, 721–724. doi:10.1038/nmat2804 (2010).
85. Friend, R. H., Gymer, R. W., Holmes, A. B., *et al.* Electroluminescence in conjugated polymers. *Nature* **397**, 121–128. doi:10.1038/16393 (1999).
86. Gray, F. M. *Polymer Electrolytes* (Royal Society of Chemistry, Cambridge, 1997).
87. Armand, M. Polymer solid electrolytes – an overview. *Solid State Ionics* **9-10**, 745–754. doi:10.1016/0167-2738(83)90083-8 (1983).
88. Pearson, R. G. Hard and Soft Acids and Bases. *Journal of the American Chemical Society* **85**, 3533–3539. doi:10.1021/ja00905a001 (1963).
89. Kim, Y., Chortos, A., Xu, W., *et al.* A bioinspired flexible organic artificial afferent nerve. *Science* **360**, 998–1003. doi:10.1126/science.aao0098 (2018).
90. Park, H.-L., Lee, Y., Kim, N., Seo, D.-G., Go, G.-T. & Lee, T.-W. Flexible Neuromorphic Electronics for Computing, Soft Robotics, and Neuroprosthetics. *Advanced Materials* **32**, 1903558. doi:10.1002/adma.201903558 (2020).

REFERENCES

91. Berthier, C., Gorecki, W., Minier, M., Armand, M. B., Chabagno, J. M. & Rigaud, P. Microscopic investigation of ionic conductivity in alkali metal salts-poly(ethylene oxide) adducts. *Solid State Ionics* **11**, 91–95. doi:10.1016/0167-2738(83)90068-1 (1983).
92. Lei, Y., Liu, Y., Xia, Y., *et al.* Memristive learning and memory functions in polyvinyl alcohol polymer memristors. *AIP Advances* **4**, 077105. doi:10.1063/1.4887010 (2014).
93. Liu, G., Wang, C., Zhang, W., *et al.* Organic Biomimicking Memristor for Information Storage and Processing Applications. *Advanced Electronic Materials* **2**, 1500298. doi:10.1002/aelm.201500298 (2016).
94. Goswami, S., Matula, A. J., Rath, S. P., *et al.* Robust resistive memory devices using solution-processable metal-coordinated azo aromatics. *Nature Materials* **16**, 1216–1224. doi:10.1038/nmat5009 (2017).
95. Goswami, S., Rath, S. P., Thompson, D., *et al.* Charge disproportionate molecular redox for discrete memristive and memcapacitive switching. *Nature Nanotechnology* **15**, 380–389. doi:10.1038/s41565-020-0653-1 (2020).
96. Zakhidov, A. A., Jung, B., Slinker, J. D., Abruña, H. D. & Malliaras, G. G. A light-emitting memristor. *Organic Electronics* **11**, 150–153. doi:10.1016/j.orgel.2009.09.015 (2010).
97. Pei, Q., Yu, G., Zhang, C., Yang, Y. & Heeger, A. J. Polymer Light-Emitting Electrochemical Cells. *Science* **269**, 1086–1088. doi:10.1126/science.269.5227.1086 (1995).
98. De Mello, J. C., Tessler, N., Graham, S. C. & Friend, R. H. Ionic space-charge effects in polymer light-emitting diodes. *Physical*

-
- Review B* **57**, 12951–12963. doi:10.1103/PhysRevB.57.12951 (1998).
99. De Mello, J. C. Interfacial feedback dynamics in polymer light-emitting electrochemical cells. *Physical Review B* **66**, 235210. doi:10.1103/PhysRevB.66.235210 (2002).
100. Matyba, P., Maturova, K., Kemerink, M., Robinson, N. D. & Edman, L. The dynamic organic p-n junction. *Nature Materials* **8**, 672–676. doi:10.1038/nmat2478 (2009).
101. Van Reenen, S., Matyba, P., Dzwilewski, A., Janssen, R. A. J., Edman, L. & Kemerink, M. A Unifying Model for the Operation of Light-Emitting Electrochemical Cells. *Journal of the American Chemical Society* **132**, 13776–13781. doi:10.1021/ja1045555 (2010).
102. Wågberg, T., Hania, P. R., Robinson, N. D., Shin, J.-H., Matyba, P. & Edman, L. On the Limited Operational Lifetime of Light-Emitting Electrochemical Cells. *Advanced Materials* **20**, 1744–1749. doi:10.1002/adma.200702595 (2008).
103. Mardegan, L., Dreessen, C., Sessolo, M., Tordera, D. & Bolink, H. J. Stable Light-Emitting Electrochemical Cells Using Hyperbranched Polymer Electrolyte. *Advanced Functional Materials* **31**, 2104249. doi:10.1002/adfm.202104249 (2021).
104. Gkoupidenis, P., Schaefer, N., Garlan, B. & Malliaras, G. G. Neuromorphic Functions in PEDOT:PSS Organic Electrochemical Transistors. *Advanced Materials* **27**, 7176–7180. doi:10.1002/adma.201503674 (2015).
105. Fang, J., Matyba, P., Robinson, N. D. & Edman, L. Identifying and alleviating electrochemical side-reactions in light-emitting electrochemical cells. *Journal of the American Chemical Society* **130**, 4562–4568. doi:10.1021/ja7113294 (2008).

REFERENCES

106. Scott, J. C. & Bozano, L. D. Nonvolatile Memory Elements Based on Organic Materials. *Advanced Materials* **19**, 1452–1463. doi:10.1002/adma.200602564 (2007).
107. Yang, Y., Ouyang, J., Ma, L., Tseng, R. J.-H. & Chu, C.-W. Electrical Switching and Bistability in Organic/Polymeric Thin Films and Memory Devices. *Advanced Functional Materials* **16**, 1001–1014. doi:10.1002/adfm.200500429 (2006).
108. Erokhin, V., Berzina, T. & Fontana, M. P. Hybrid electronic device based on polyaniline-polyethyleneoxide junction. *Journal of Applied Physics* **97**, 064501. doi:10.1063/1.1861508 (2005).
109. Erokhin, V., Berzina, T., Camorani, P. & Fontana, M. P. Non-equilibrium electrical behaviour of polymeric electrochemical junctions. *Journal of Physics: Condensed Matter* **19**, 205111. doi:10.1088/0953-8984/19/20/205111 (2007).
110. Smerieri, A., Berzina, T., Erokhin, V. & Fontana, M. P. Polymeric electrochemical element for adaptive networks: Pulse mode. *Journal of Applied Physics* **104**, 114513. doi:10.1063/1.3033399 (2008).
111. Berzina, T., Smerieri, A., Bernabò, M., *et al.* Optimization of an organic memristor as an adaptive memory element. *Journal of Applied Physics* **105**, 124515. doi:10.1063/1.3153944 (2009).
112. Gkoupidenis, P., Koutsouras, D. A. & Malliaras, G. G. Neuromorphic device architectures with global connectivity through electrolyte gating. *Nature Communications* **8**, 15448. doi:10.1038/ncomms15448 (2017).
113. Xu, W., Min, S.-Y., Hwang, H. & Lee, T.-W. Organic core-sheath nanowire artificial synapses with femtojoule energy consumption. *Science Advances* **2**, e1501326. doi:10.1126/sciadv.1501326 (2016).

-
114. Seo, D.-G., Lee, Y., Go, G.-T., *et al.* Versatile neuromorphic electronics by modulating synaptic decay of single organic synaptic transistor: From artificial neural networks to neuro-prosthetics. *Nano Energy* **65**, 104035. doi:10.1016/j.nanoen.2019.104035 (2019).
115. Heremans, P., Gelinck, G. H., Müller, R., Baeg, K.-J., Kim, D.-Y. & Noh, Y.-Y. Polymer and Organic Nonvolatile Memory Devices. *Chemistry of Materials* **23**, 341–358. doi:10.1021/cm102006v (2011).
116. Yang, M., Zhao, X., Tang, Q., *et al.* Stretchable and conformable synapse memristors for wearable and implantable electronics. *Nanoscale* **10**, 18135–18144. doi:10.1039/C8NR05336G (2018).
117. Kim, M.-K. & Lee, J.-S. Ultralow Power Consumption Flexible Biomemristors. *ACS Applied Materials & Interfaces* **10**, 10280–10286. doi:10.1021/acsami.8b01781 (2018).
118. Shih, C.-C., Lee, W.-Y. & Chen, W.-C. Nanostructured materials for non-volatile organic transistor memory applications. *Materials Horizons* **3**, 294–308. doi:10.1039/C6MH00049E (2016).
119. Marković, D., Mizrahi, A., Querlioz, D. & Grollier, J. Physics for neuromorphic computing. *Nature Reviews Physics* **2**, 499–510. doi:10.1038/s42254-020-0208-2 (2020).
120. Harikesh, P. C., Tu, D. & Fabiano, S. Organic electrochemical neurons for neuromorphic perception. *Nature Electronics* **7**, 525–536. doi:10.1038/s41928-024-01200-5 (2024).
121. Krauhausen, I., Griggs, S., McCulloch, I., den Toonder, J. M. J., Gkoupidenis, P., van de Burgt, Y., *et al.* Bio-inspired multimodal learning with organic neuromorphic electronics for behavioral conditioning in robotics. *Nature Communications* **15**, 4765. doi:10.1038/s41467-024-48881-2 (2024).

REFERENCES

122. Xia, Y., Zhang, C., Xu, Z., *et al.* Organic iontronic memristors for artificial synapses and bionic neuromorphic computing. *Nanoscale* **16**, 1471–1489. doi:10.1039/D3NR06057H (2024).
123. Fenton, D. E., Parker, J. M. & Wright, P. V. Complexes of alkali metal ions with poly(ethylene oxide). *Polymer* **14**, 589. doi:10.1016/0032-3861(73)90146-8 (1973).
124. Wright, P. V. Electrical conductivity in ionic complexes of poly(ethylene oxide). *British Polymer Journal* **7**, 319–327. doi:10.1002/pi.4980070505 (1975).
125. Lightfoot, P., Mehta, M. A. & Bruce, P. G. Crystal Structure of the Polymer Electrolyte Poly(ethylene oxide)₃:LiCF₃SO₃. *Science* **262**, 883–885. doi:10.1126/science.262.5135.883 (1993).
126. Froehling, P. Development of DSM's Hybrane hyperbranched polyesteramides. *Journal of Polymer Science Part A: Polymer Chemistry* **42**, 3110–3115. doi:10.1002/pola.20113 (2004).
127. Bruce, P. G. & Vincent, C. A. Polymer electrolytes. *Journal of the Chemical Society, Faraday Transactions* **89**, 3187–3203. doi:10.1039/ft9938903187 (1993).
128. Kreuer, K.-D., Rabenau, A. & Weppner, W. Vehicle Mechanism, A New Model for the Interpretation of the Conductivity of Fast Proton Conductors. *Angewandte Chemie International Edition in English* **21**, 208–209. doi:10.1002/anie.198202082 (1982).
129. Agmon, N. The Grotthuss mechanism. *Chemical Physics Letters* **244**, 456–462. doi:10.1016/0009-2614(95)00905-J (1995).
130. Druger, S. D., Nitzan, A. & Ratner, M. A. Dynamic bond percolation theory: A microscopic model for diffusion in dynamically disordered systems. I. Definition and one-dimensional case. *The Journal of Chemical Physics* **79**, 3133–3142. doi:10.1063/1.446144 (1983).

-
131. Ratner, M. A. & Shriver, D. F. Ion transport in solvent-free polymers. *Chemical Reviews* **88**, 109–124. doi:10.1021/cr00083a006 (1988).
132. Angell, C. A. Relaxation in liquids, polymers and plastic crystals — strong/fragile patterns and problems. *Journal of Non-Crystalline Solids* **131–133**, 13–31. doi:10.1016/0022-3093(91)90266-9 (1991).
133. Gadjourova, Z., Andreev, Y. G., Tunstall, D. P. & Bruce, P. G. Ionic conductivity in crystalline polymer electrolytes. *Nature* **412**, 520–523. doi:10.1038/35087538 (2001).
134. Voit, B. I. & Lederer, A. Hyperbranched and Highly Branched Polymer Architectures—Synthetic Strategies and Major Characterization Aspects. *Chemical Reviews* **109**, 5924–5973. doi:10.1021/cr900068q (2009).
135. Tang, S. & Edman, L. Light-Emitting Electrochemical Cells: A Review on Recent Progress. *Topics in Current Chemistry* **374**, 40. doi:10.1007/s41061-016-0040-4 (2016).
136. Smith, D. L. Steady state model for polymer light-emitting electrochemical cells. *Journal of Applied Physics* **81**, 2869–2880. doi:10.1063/1.363966 (1997).
137. Sandström, A., Dam, H. F., Krebs, F. C. & Edman, L. Ambient fabrication of flexible and large-area organic light-emitting devices using slot-die coating. *Nature Communications* **3**, 1002. doi:10.1038/ncomms2002 (2012).
138. Hohertz, D. & Gao, J. How Electrode Work Function Affects Doping and Electroluminescence of Polymer Light-Emitting Electrochemical Cells. *Advanced Materials* **20**, 3298–3302. doi:10.1002/adma.200800068 (2008).

REFERENCES

139. AlTal, F. & Gao, J. Long-term testing of polymer light-emitting electrochemical cells: Reversible doping and black spots. *Organic Electronics* **18**, 1–7. doi:10.1016/j.orgel.2014.12.028 (2015).
140. Tordera, D., Frey, J., Vonlanthen, D., *et al.* Low Current Density Driving Leads to Efficient, Bright and Stable Green Electroluminescence. *Advanced Energy Materials* **3**, 1338–1343. doi:10.1002/aenm.201300284 (2013).